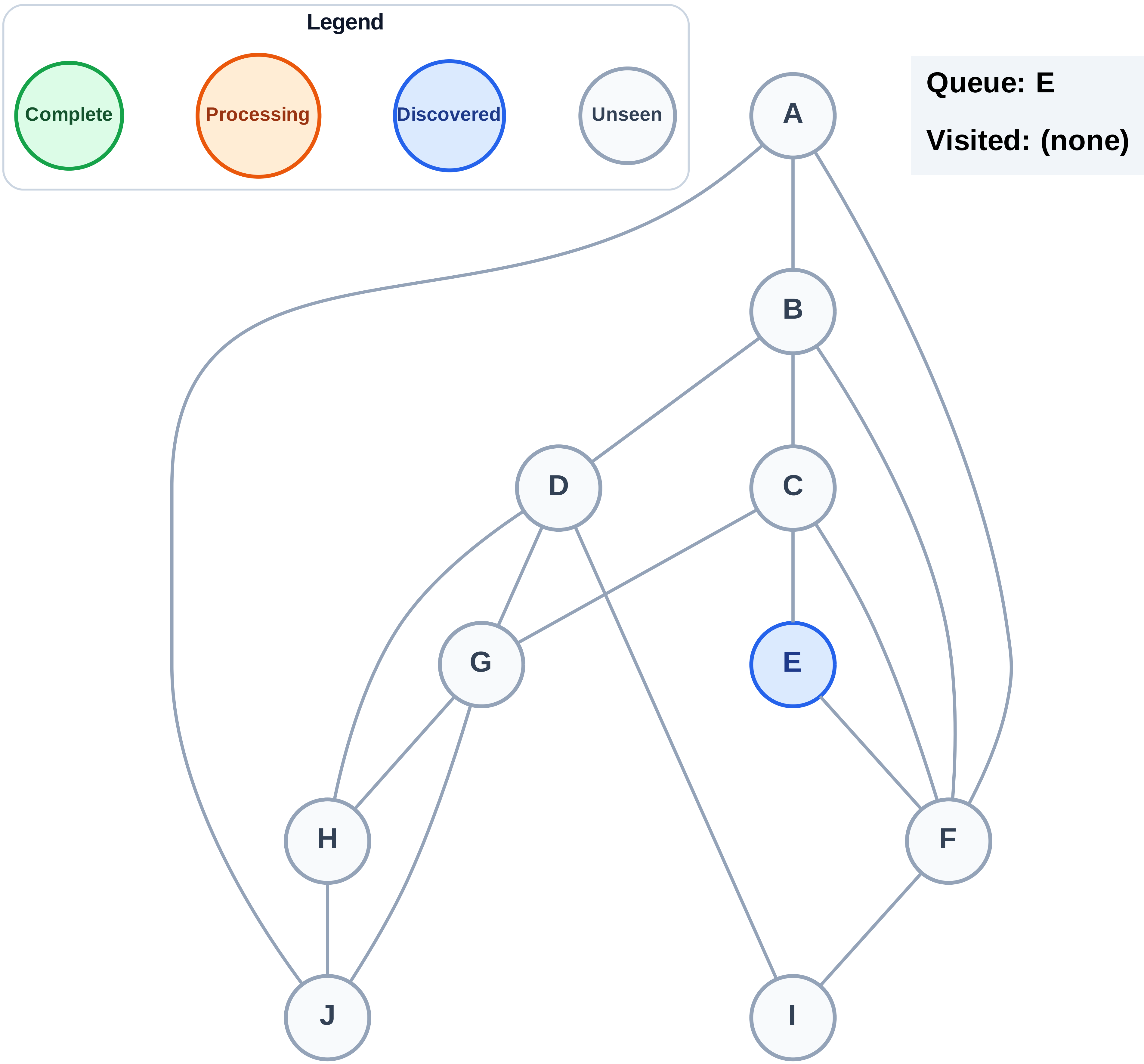
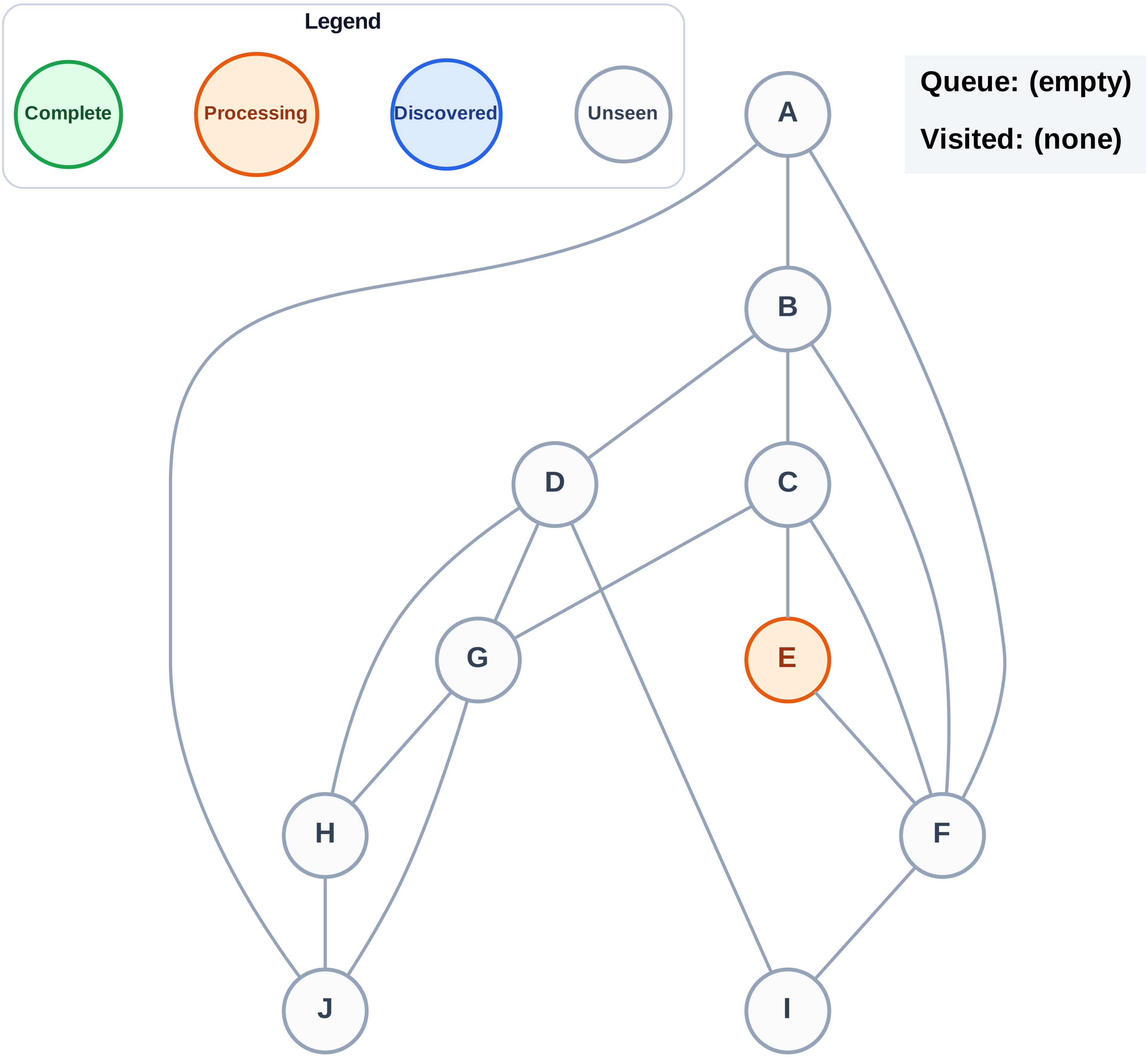
# BFS Traversal

Step 1: Start at E



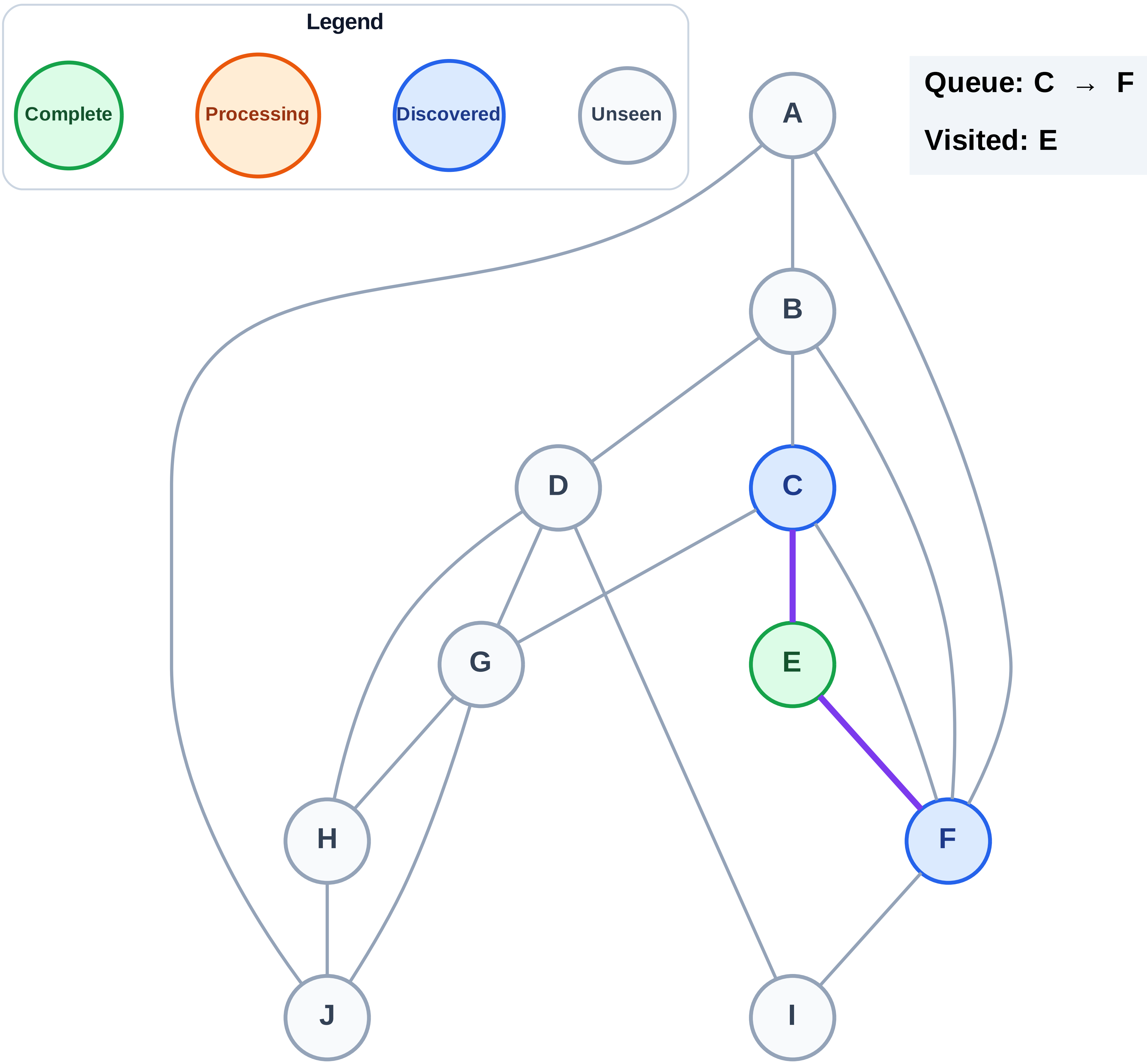
# BFS Traversal

Step 2: Process node E



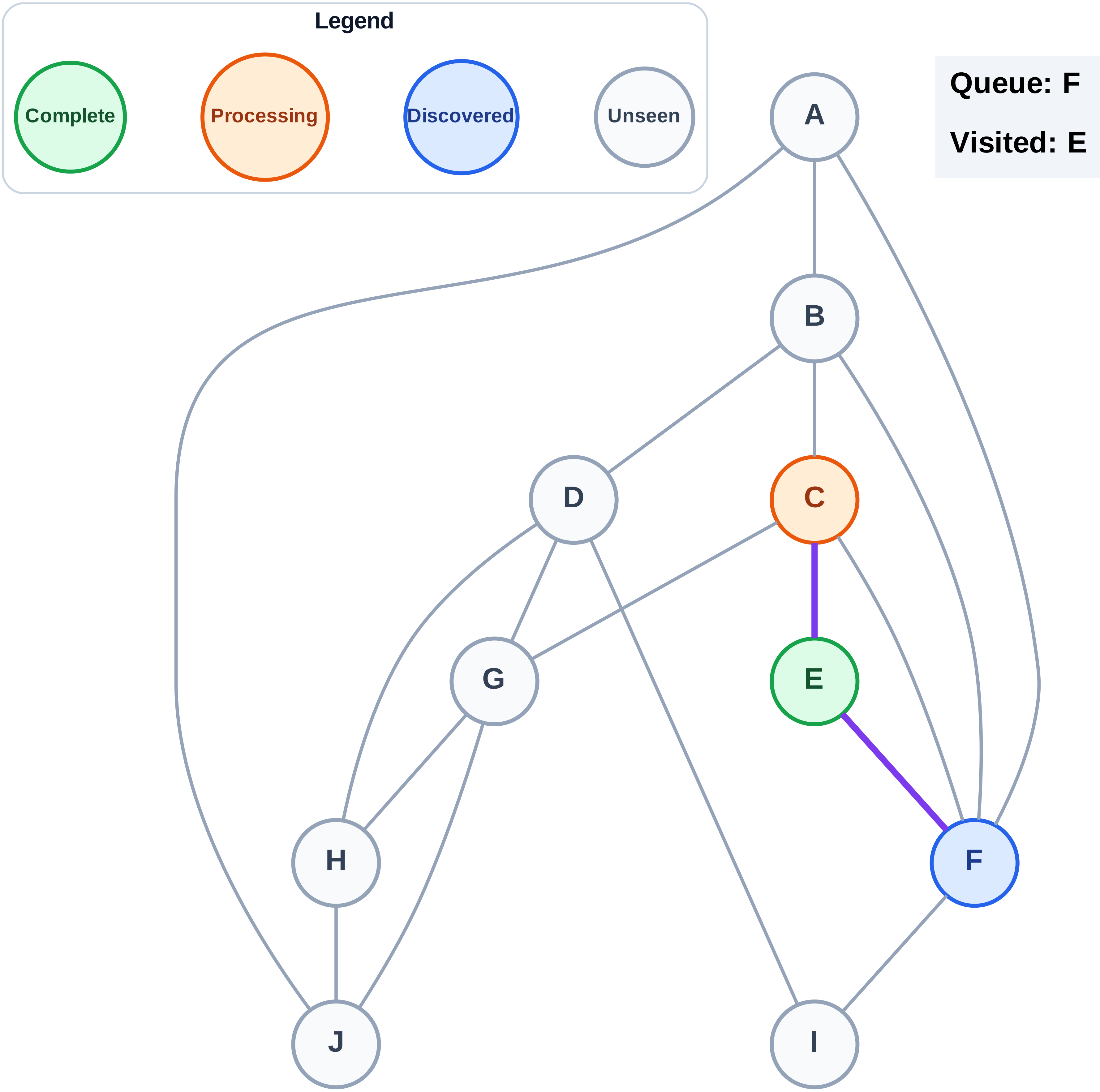
# BFS Traversal

Step 3: Discover C, F from E



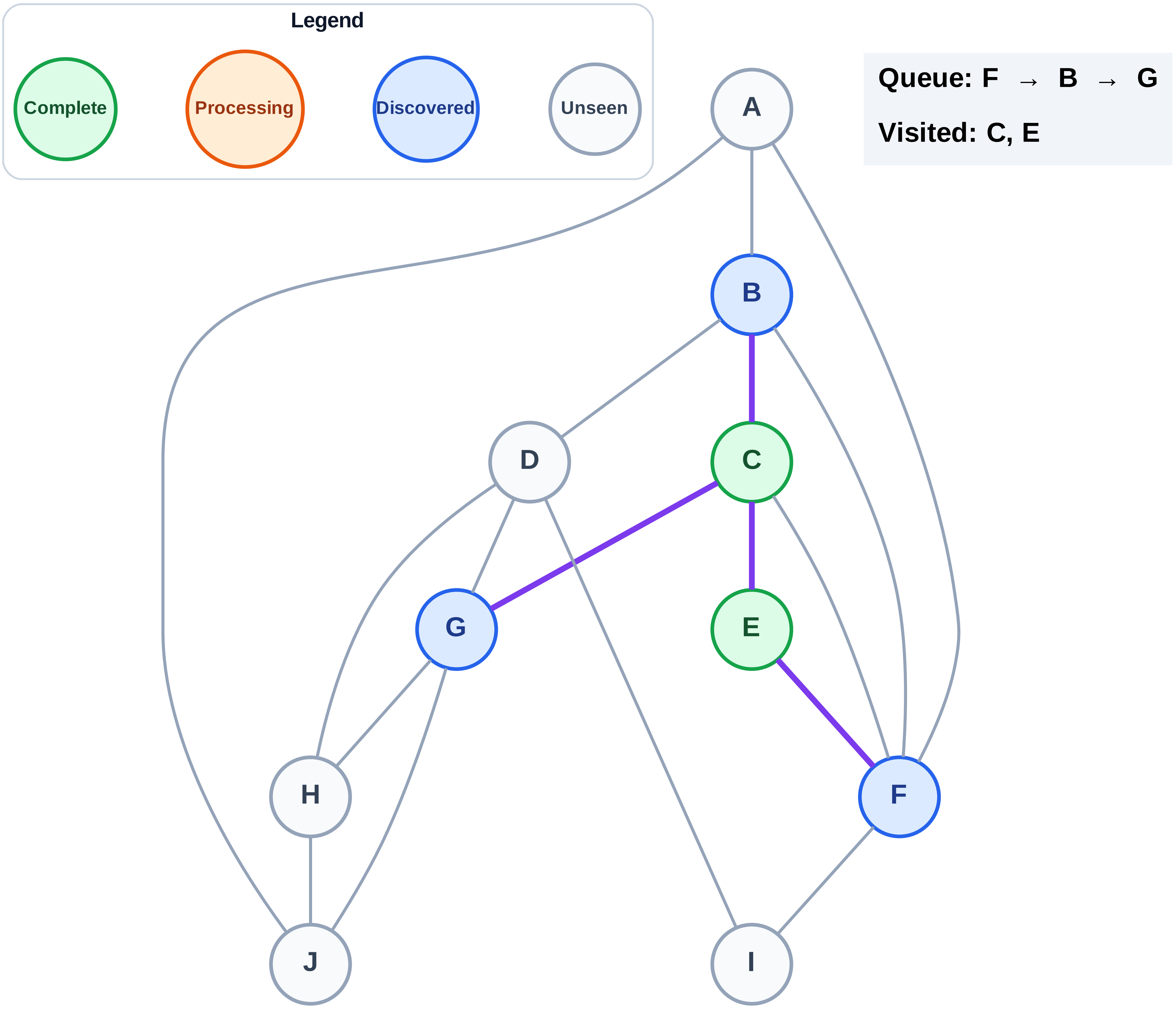
# BFS Traversal

Step 4: Process node C



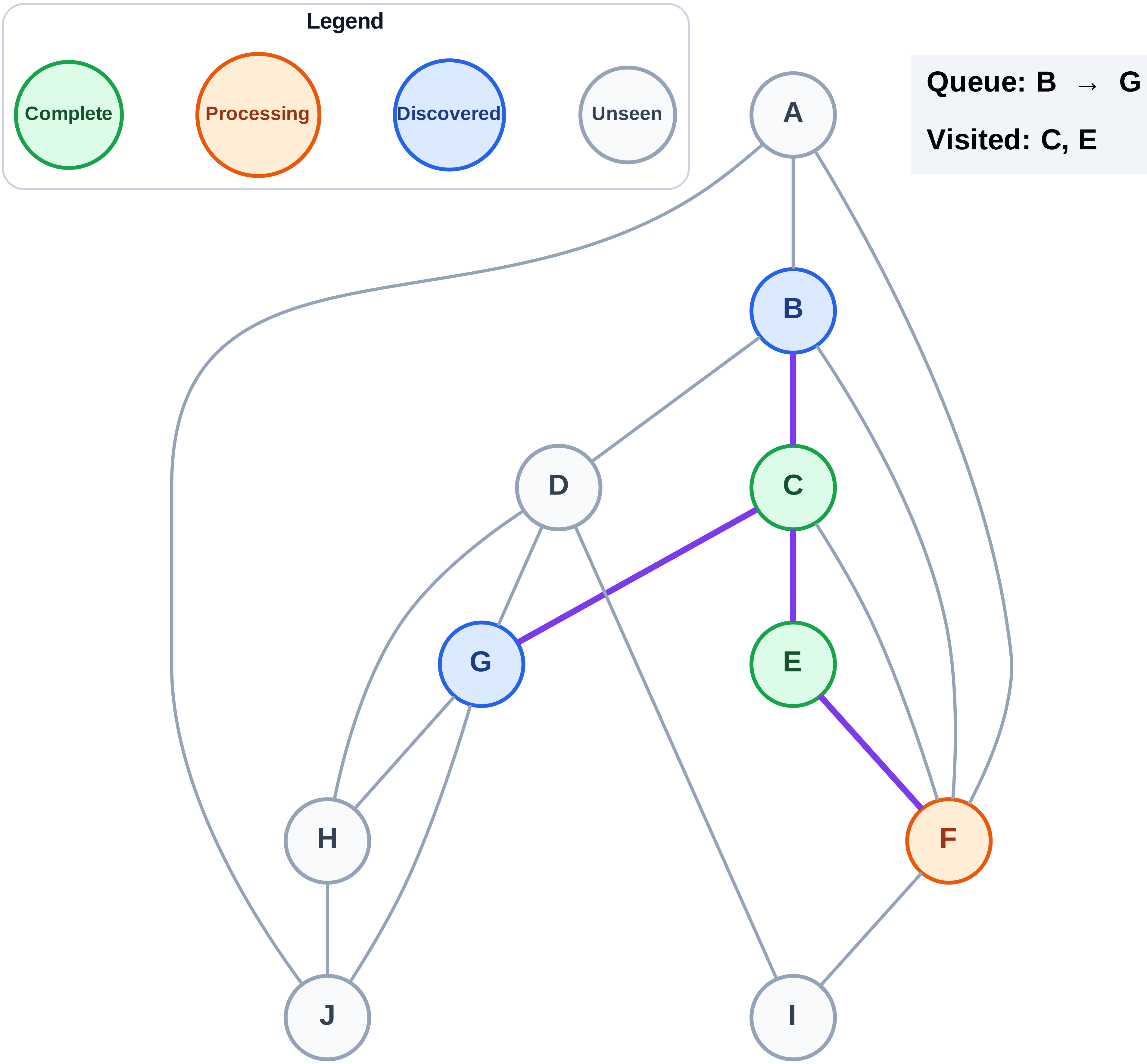
# BFS Traversal

Step 5: Discover B, G from C



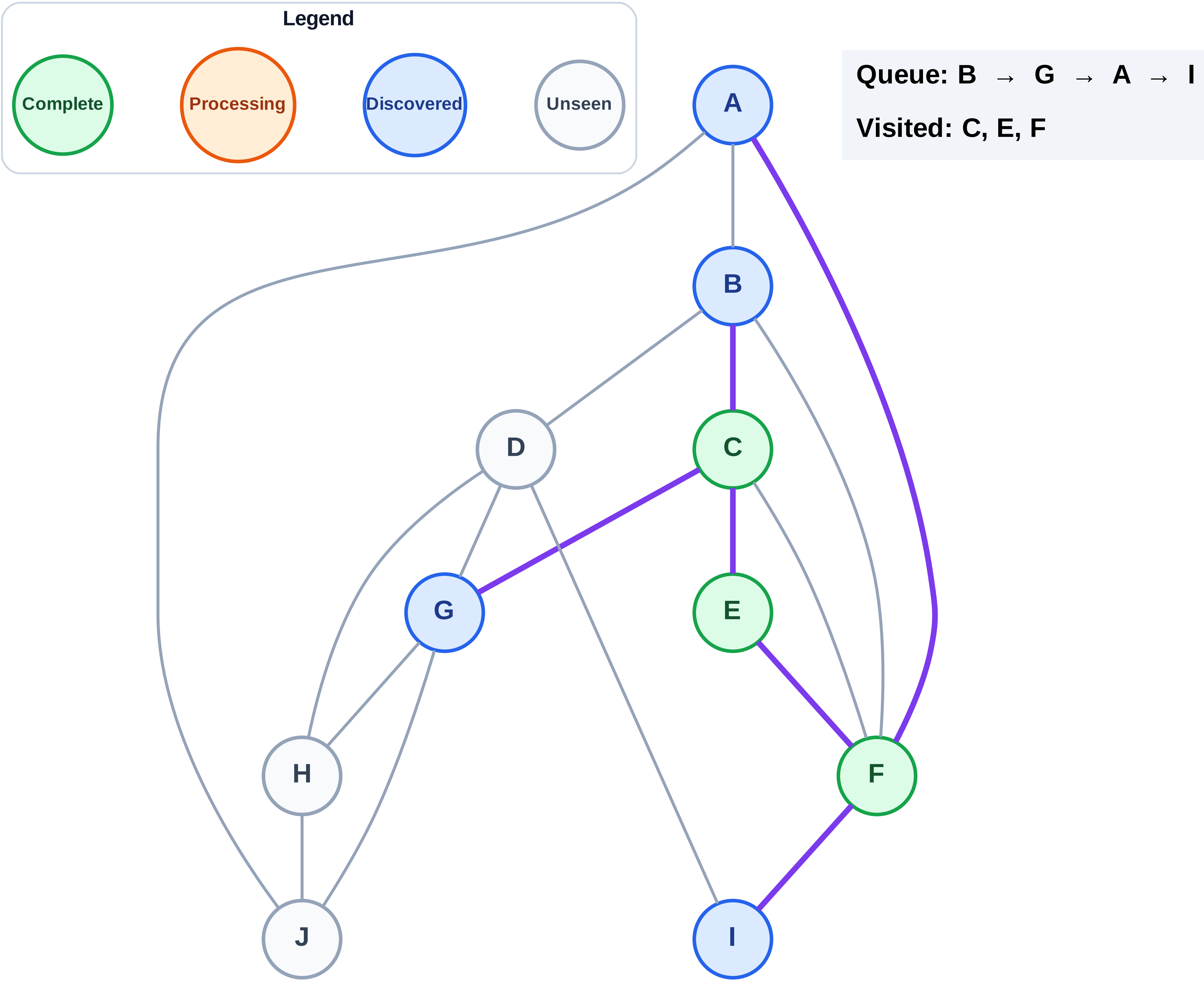
# BFS Traversal

Step 6: Process node F



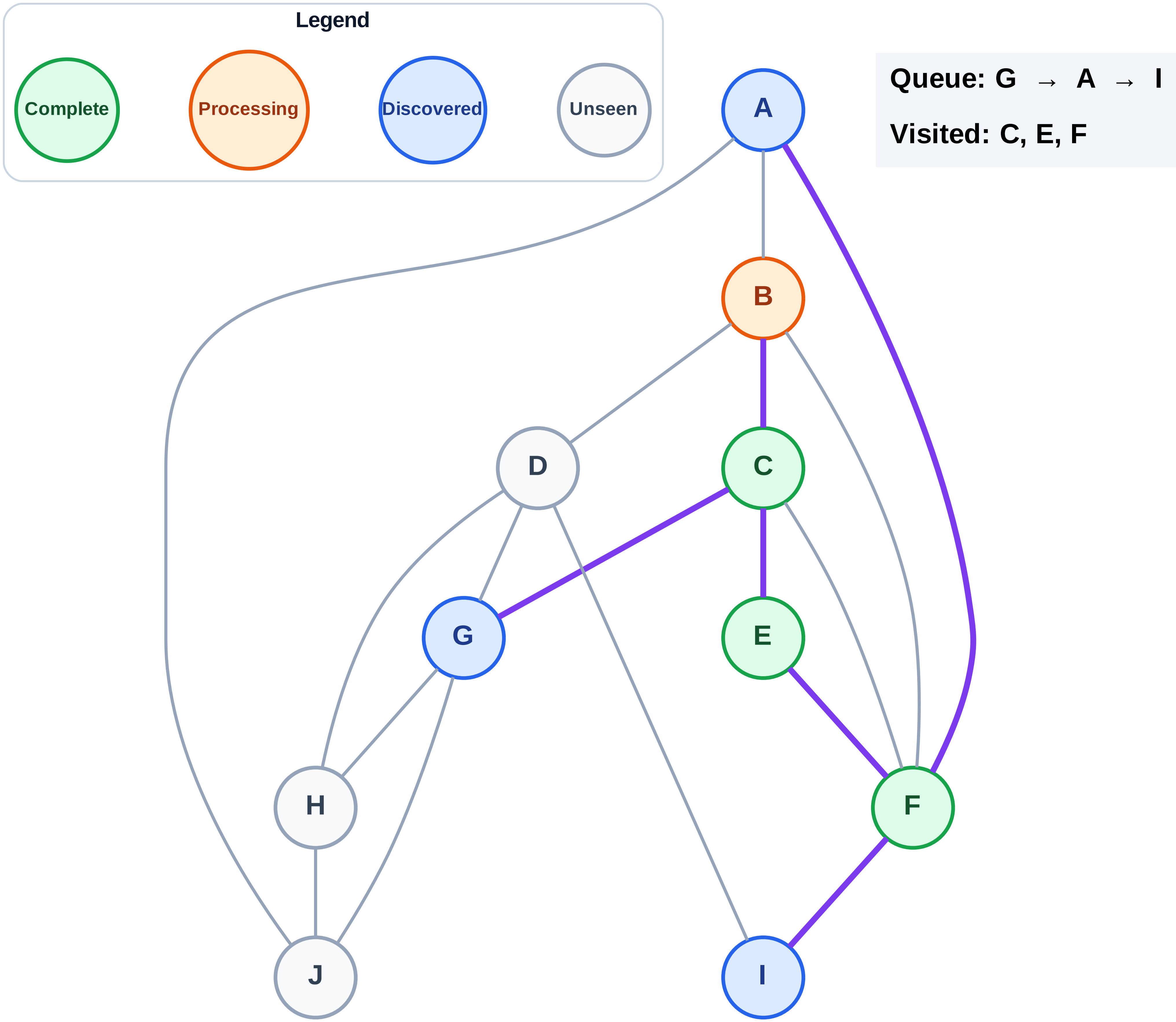
# BFS Traversal

Step 7: Discover A, I from F



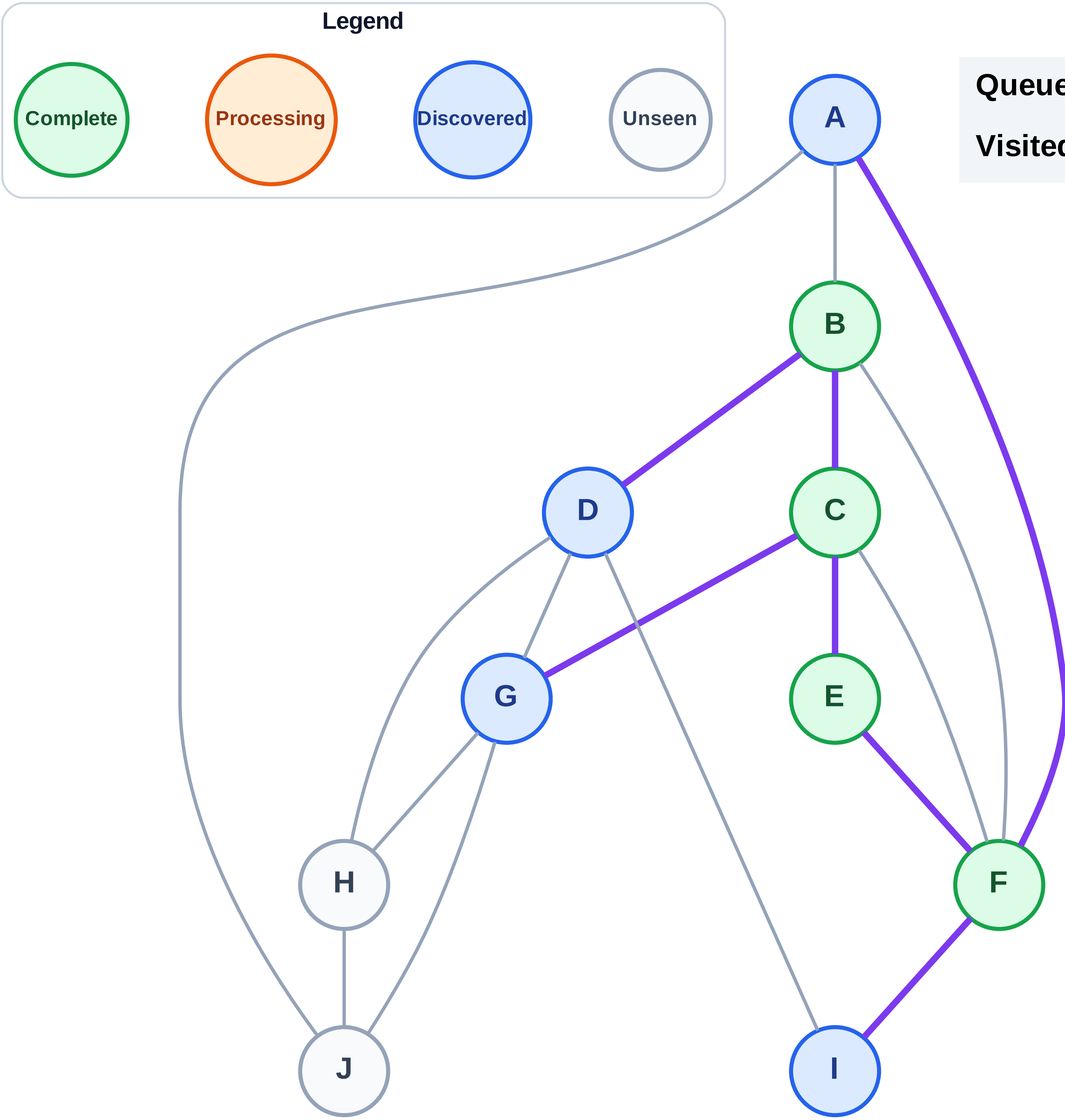
# BFS Traversal

Step 8: Process node B



# BFS Traversal

Step 9: Discover D from B

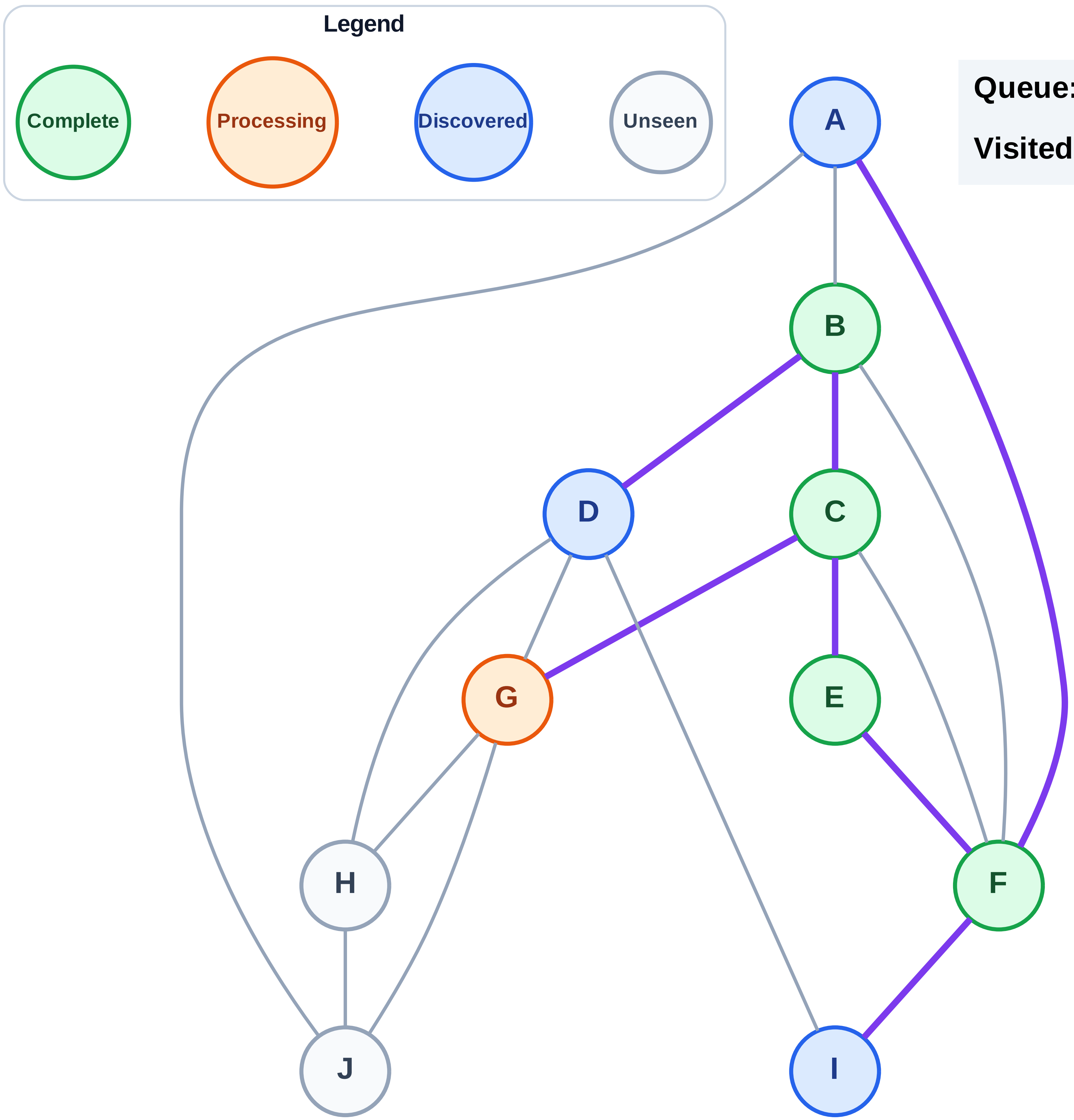



**Queue:** G → A → I → D

**Visited:** B, C, E, F

# BFS Traversal

Step 10: Process node G

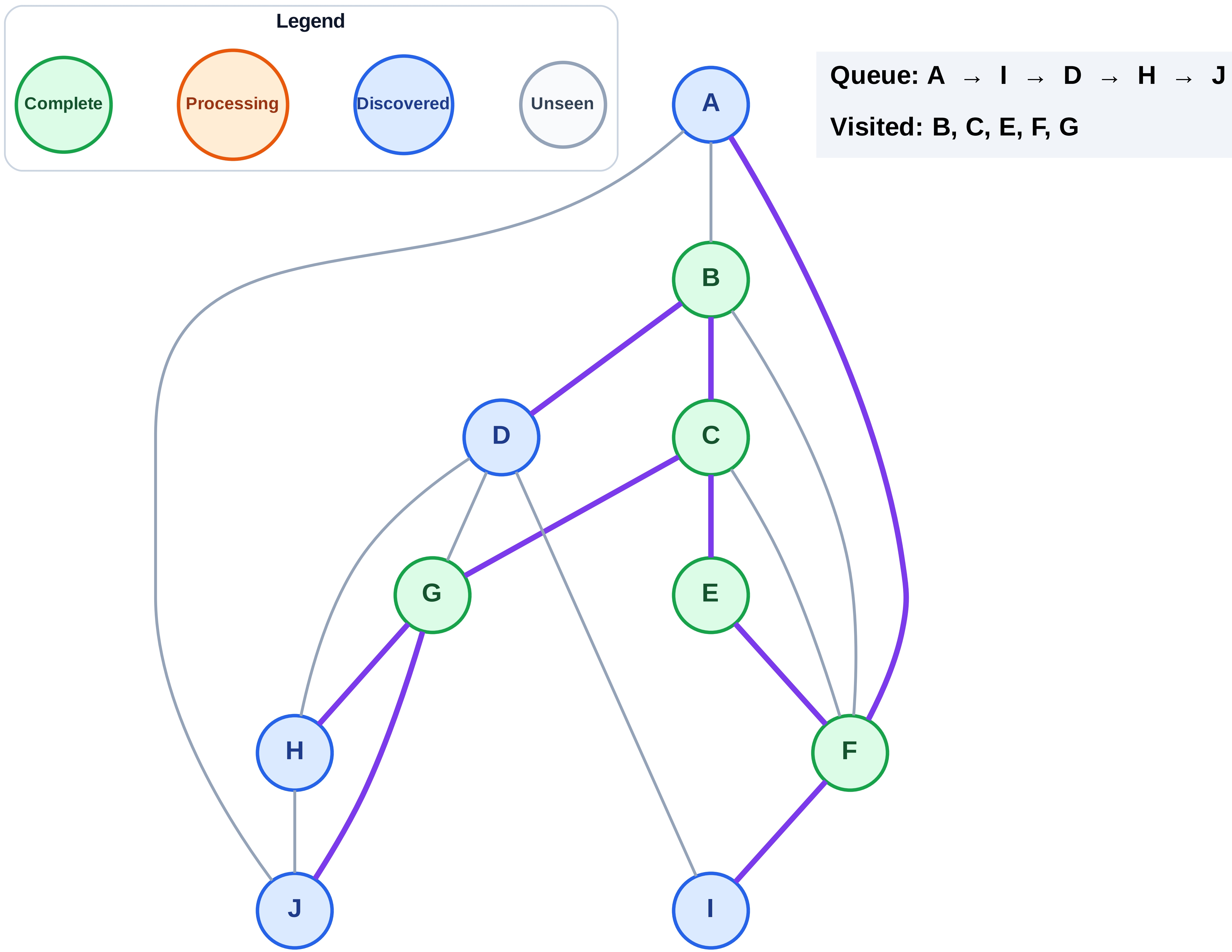



**Queue:** A → I → D

**Visited:** B, C, E, F

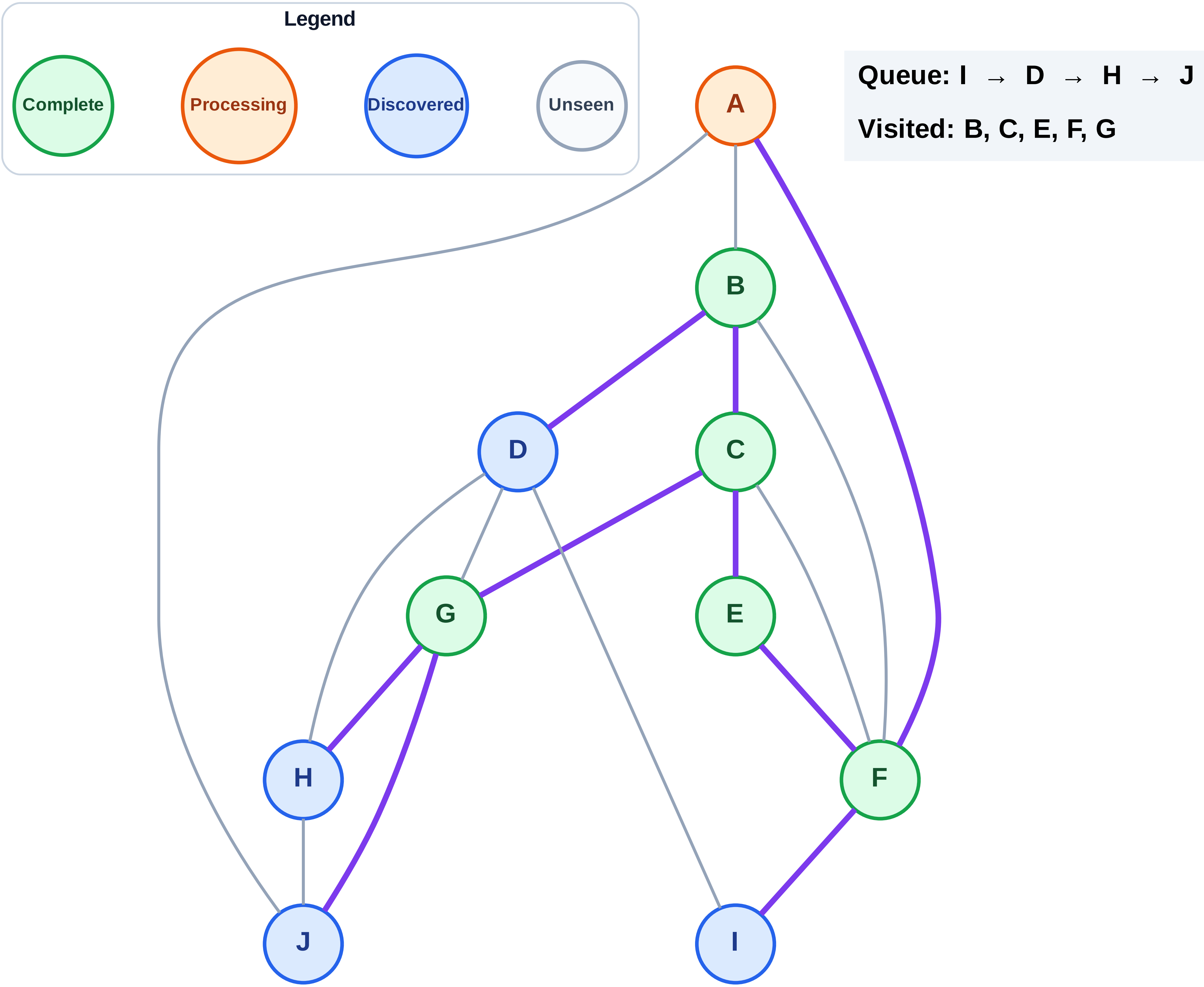
# BFS Traversal

Step 11: Discover H, J from G



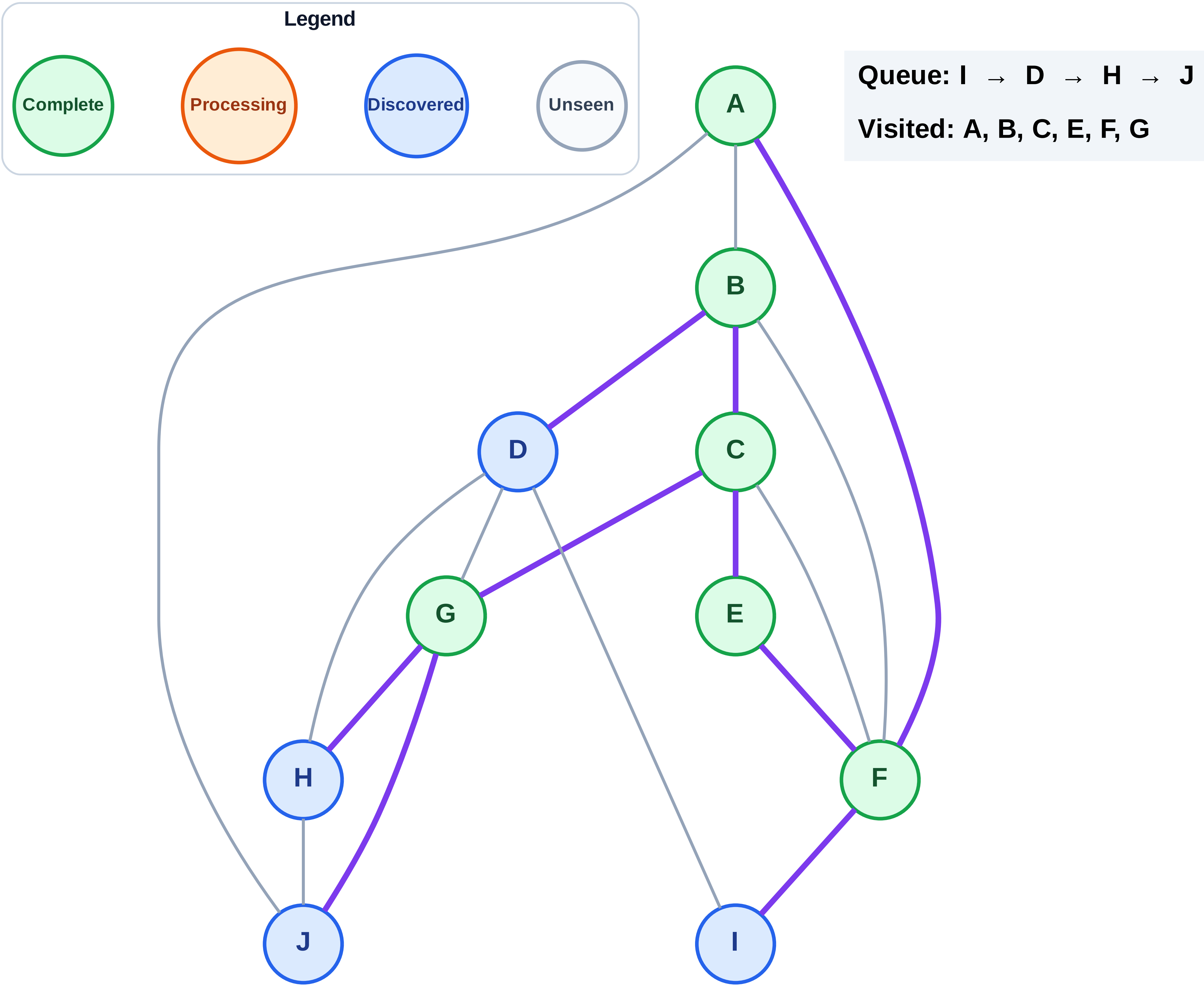
# BFS Traversal

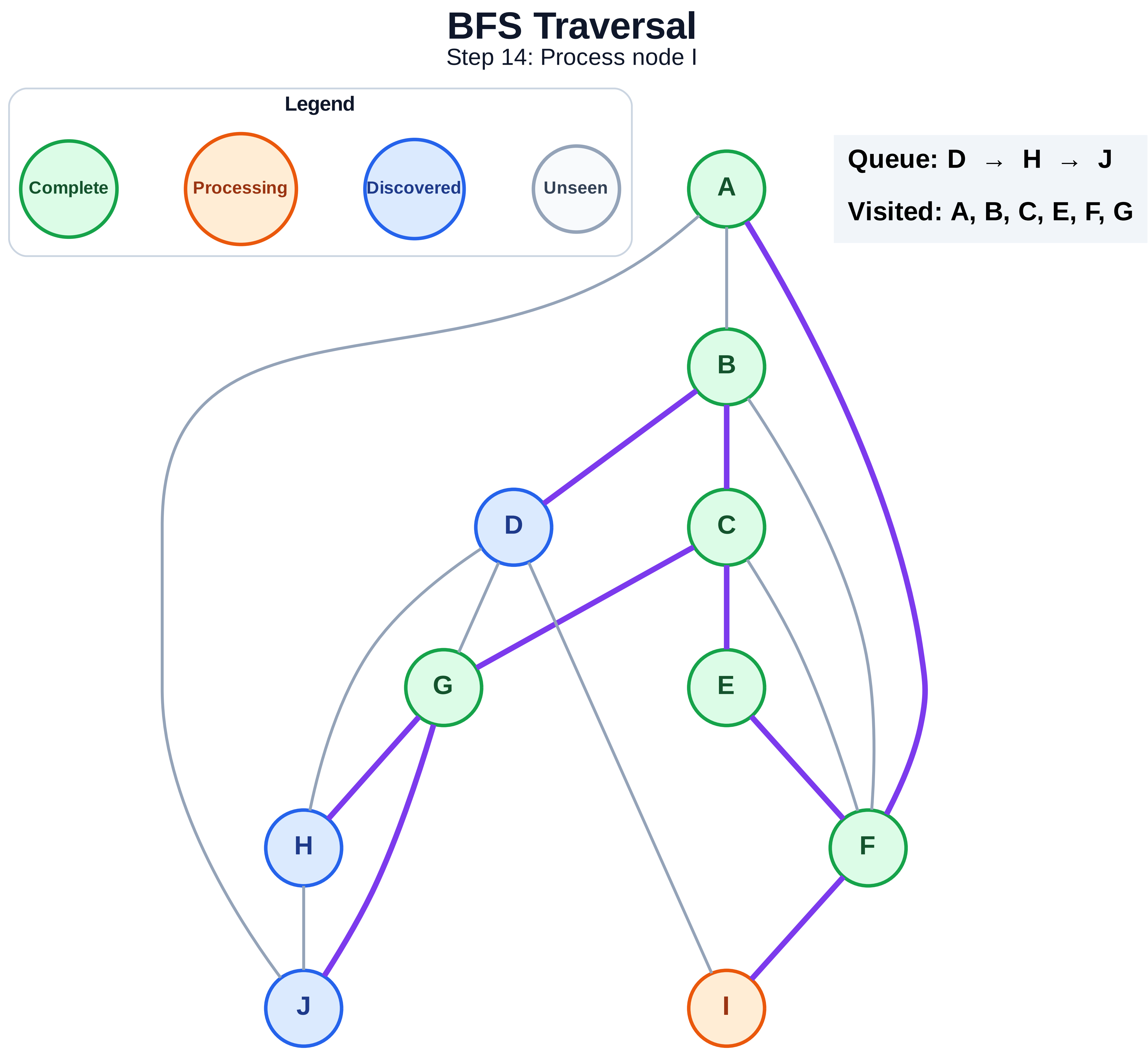
Step 12: Process node A

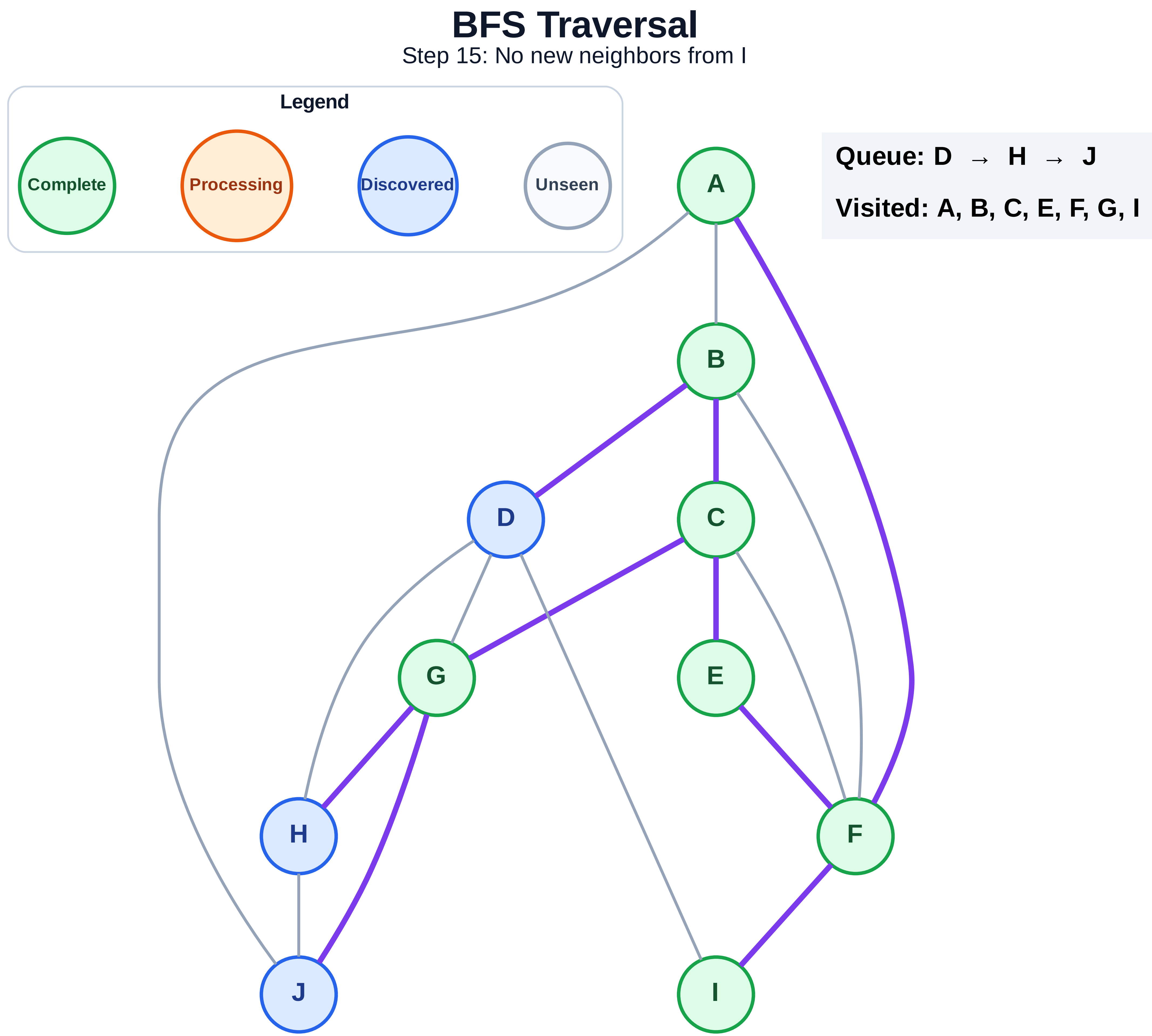


# BFS Traversal

Step 13: No new neighbors from A

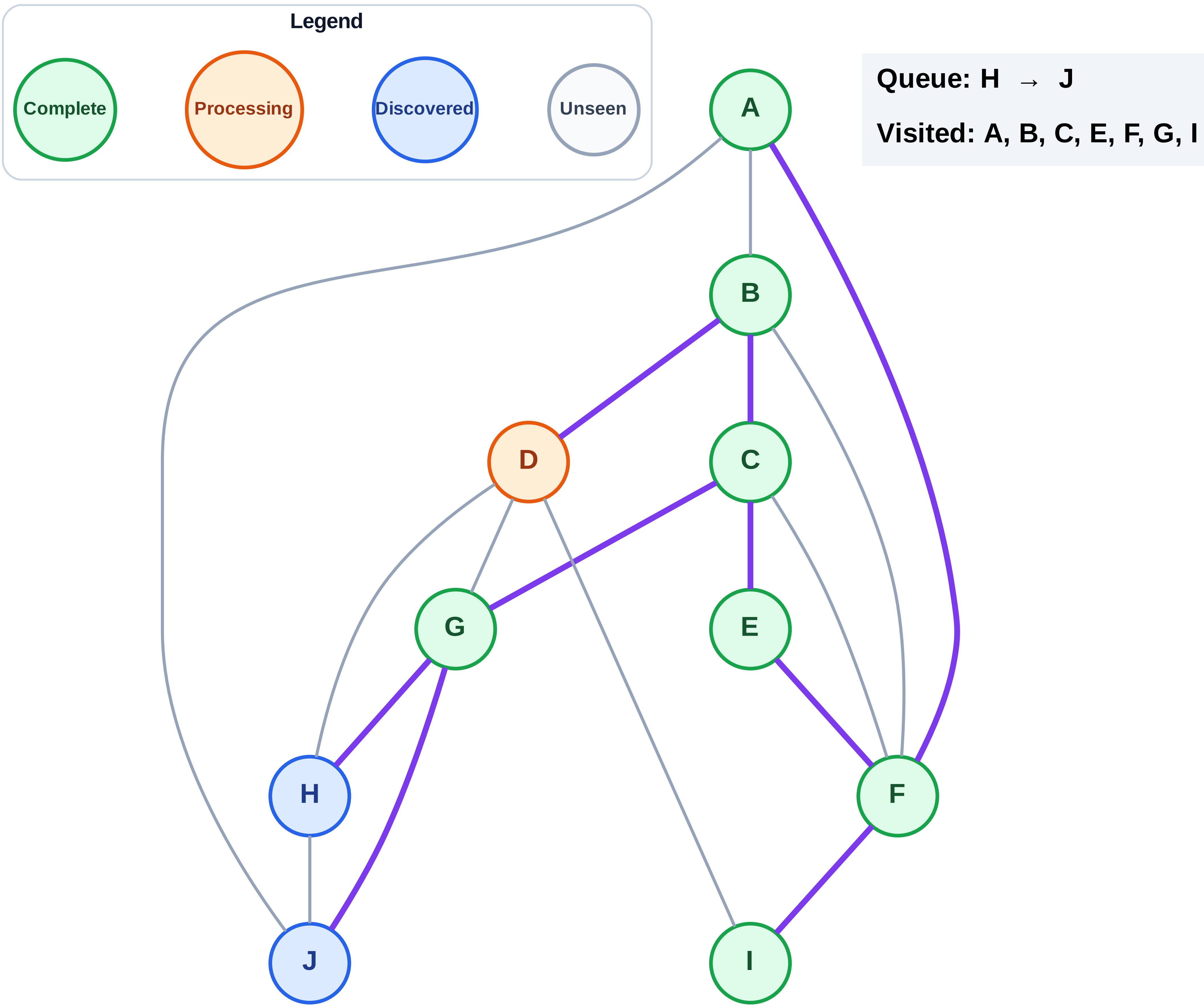



BFS Traversal
Step 14: Process node I
Legend
Complete
Processing
Discovered
Unseen
Queue: D → H → J
Visited: A, B, C, E, F, G
A
B
C
D
E
F
G
H
I
J


BFS Traversal
Step 15: No new neighbors from I
Legend
Complete
Processing
Discovered
Unseen
Queue: D → H → J
Visited: A, B, C, E, F, G, I
A
B
C
D
E
F
G
H
I
J

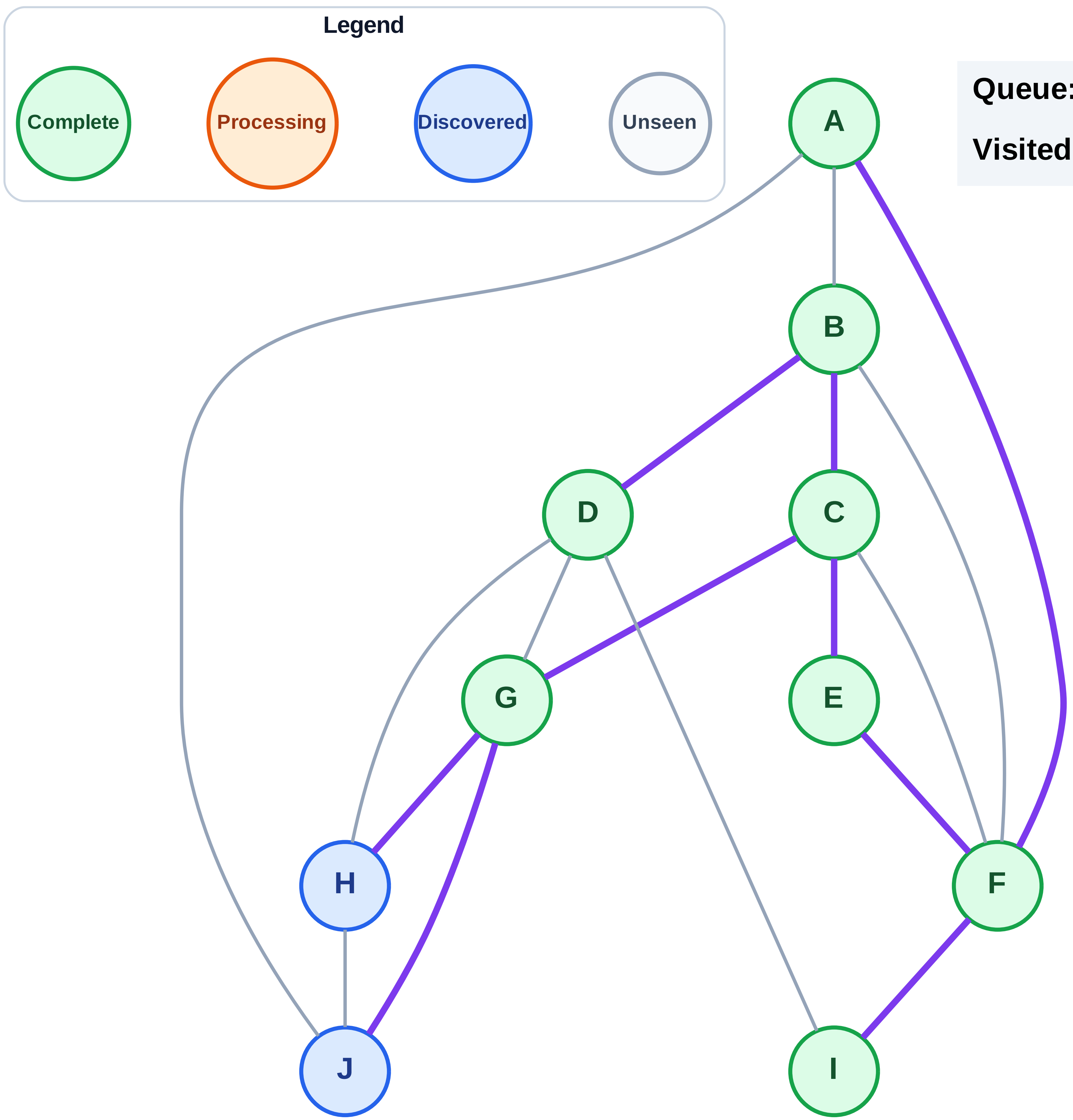
# BFS Traversal

Step 16: Process node D



# BFS Traversal

Step 17: No new neighbors from D

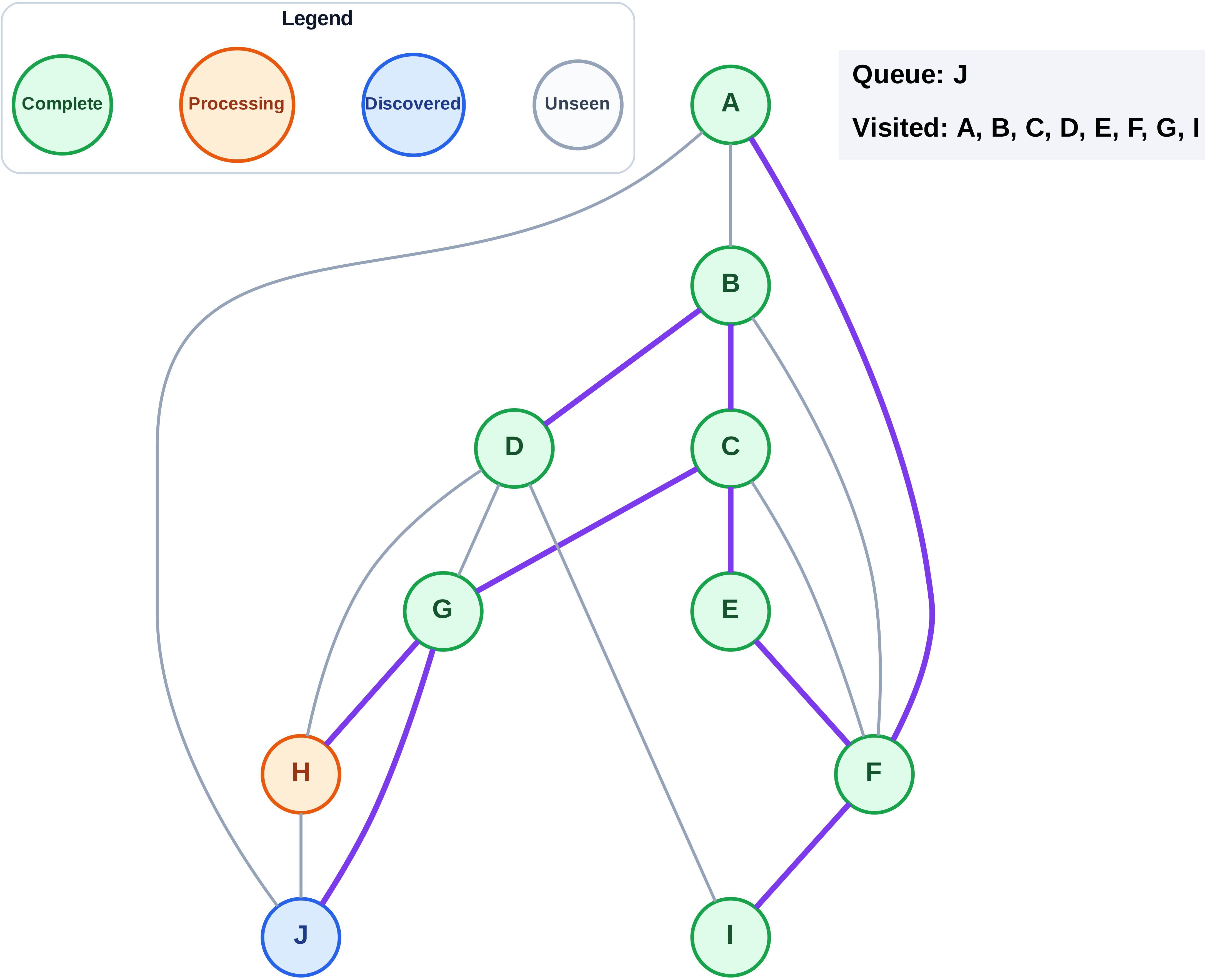


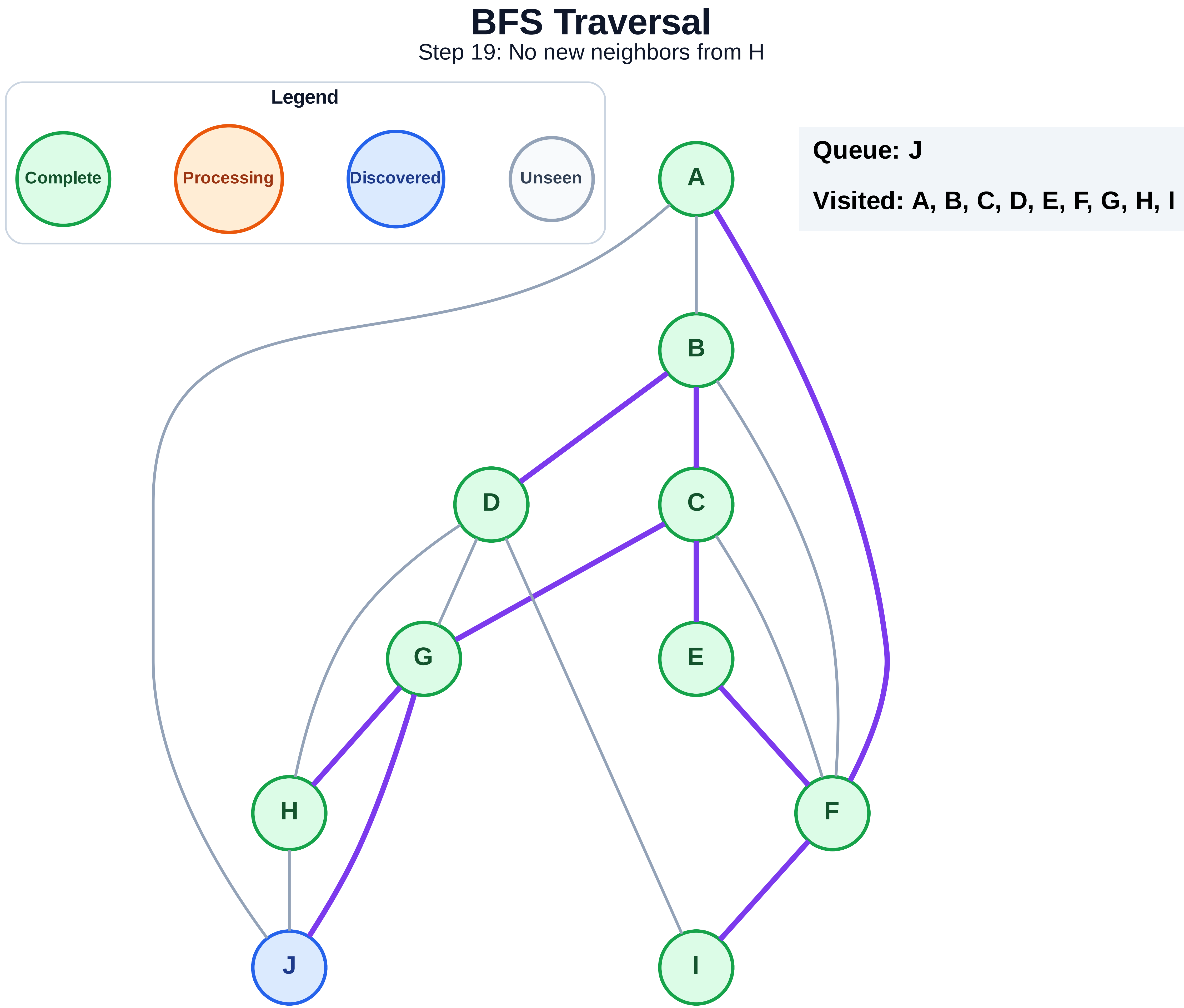

**Queue: H → J**

**Visited: A, B, C, D, E, F, G, I**

# BFS Traversal

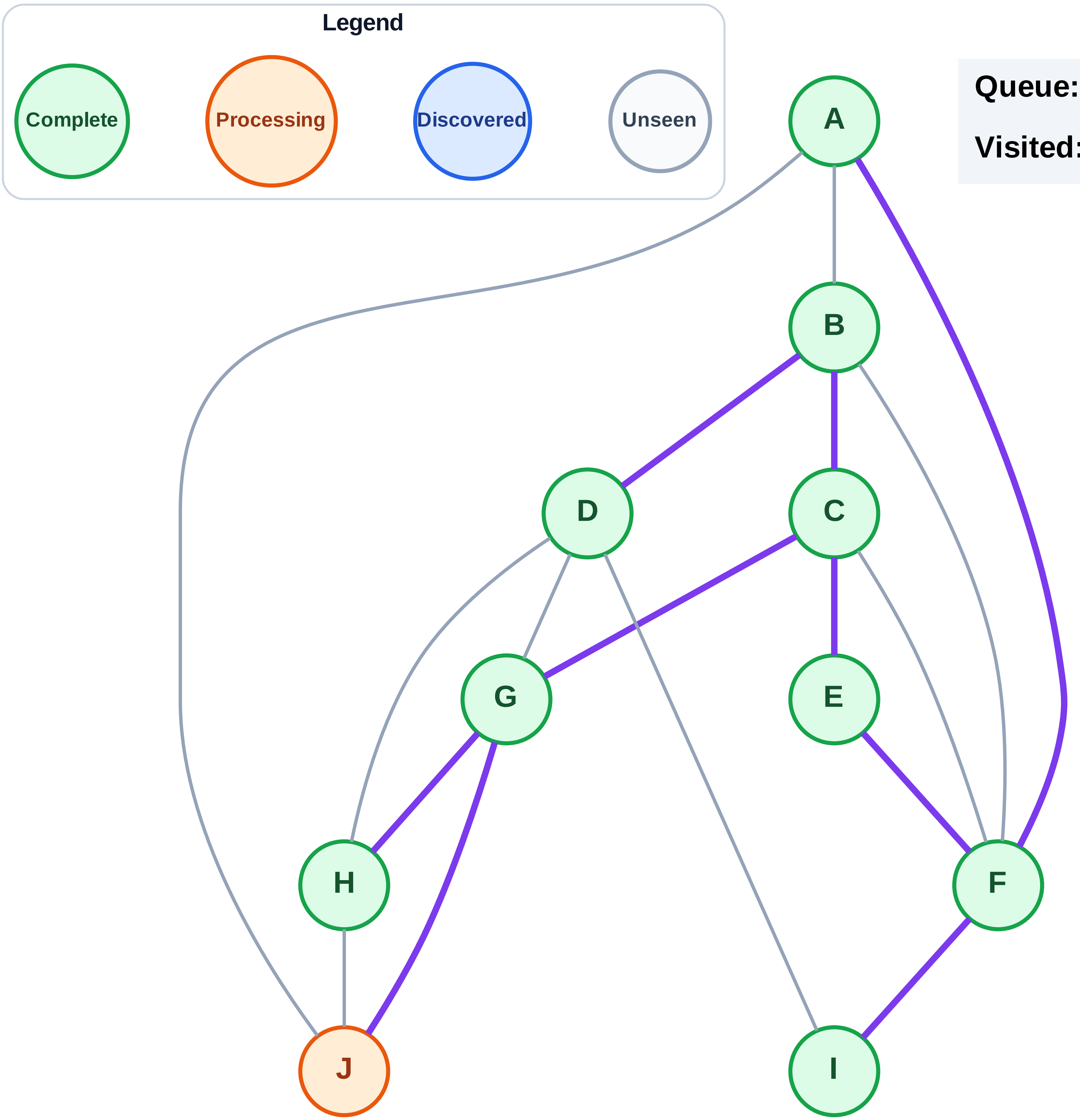
Step 18: Process node H




BFS Traversal
Step 19: No new neighbors from H
Legend
Complete
Processing
Discovered
Unseen
Queue: J
Visited: A, B, C, D, E, F, G, H, I
A
B
C
D
E
F
G
H
I
J

# BFS Traversal

Step 20: Process node J

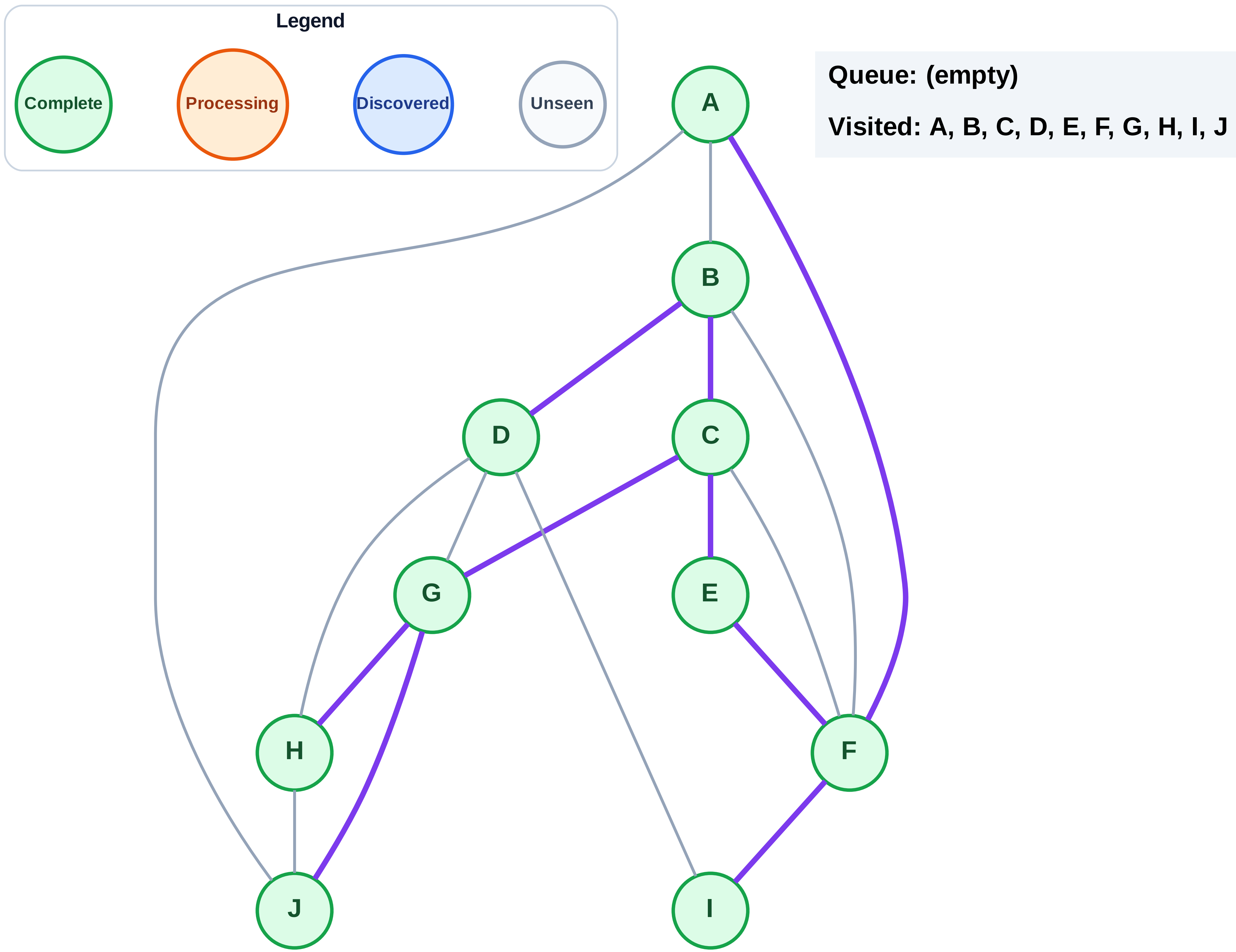



**Queue: (empty)**

**Visited: A, B, C, D, E, F, G, H, I**

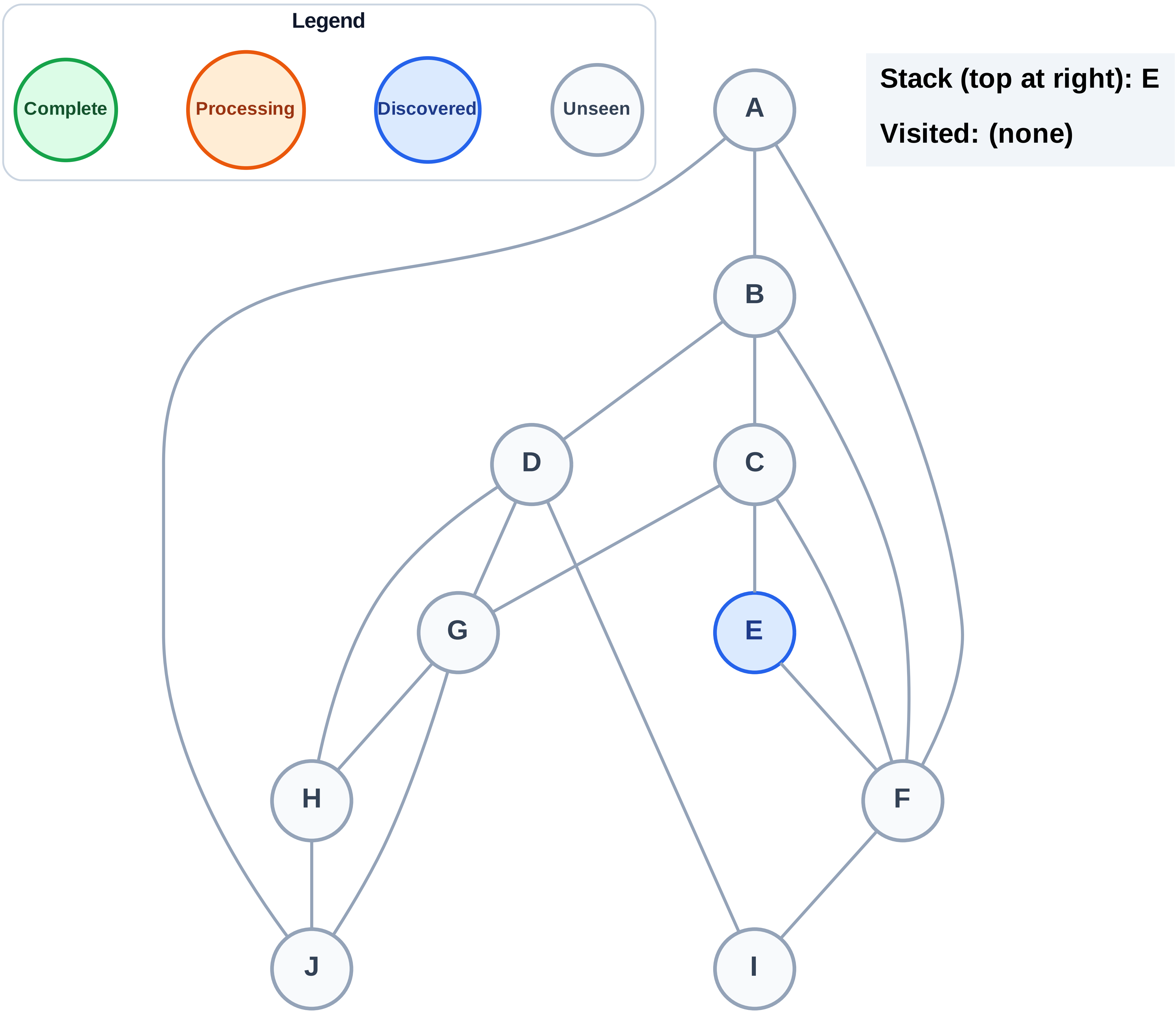
# BFS Traversal

Step 21: No new neighbors from J



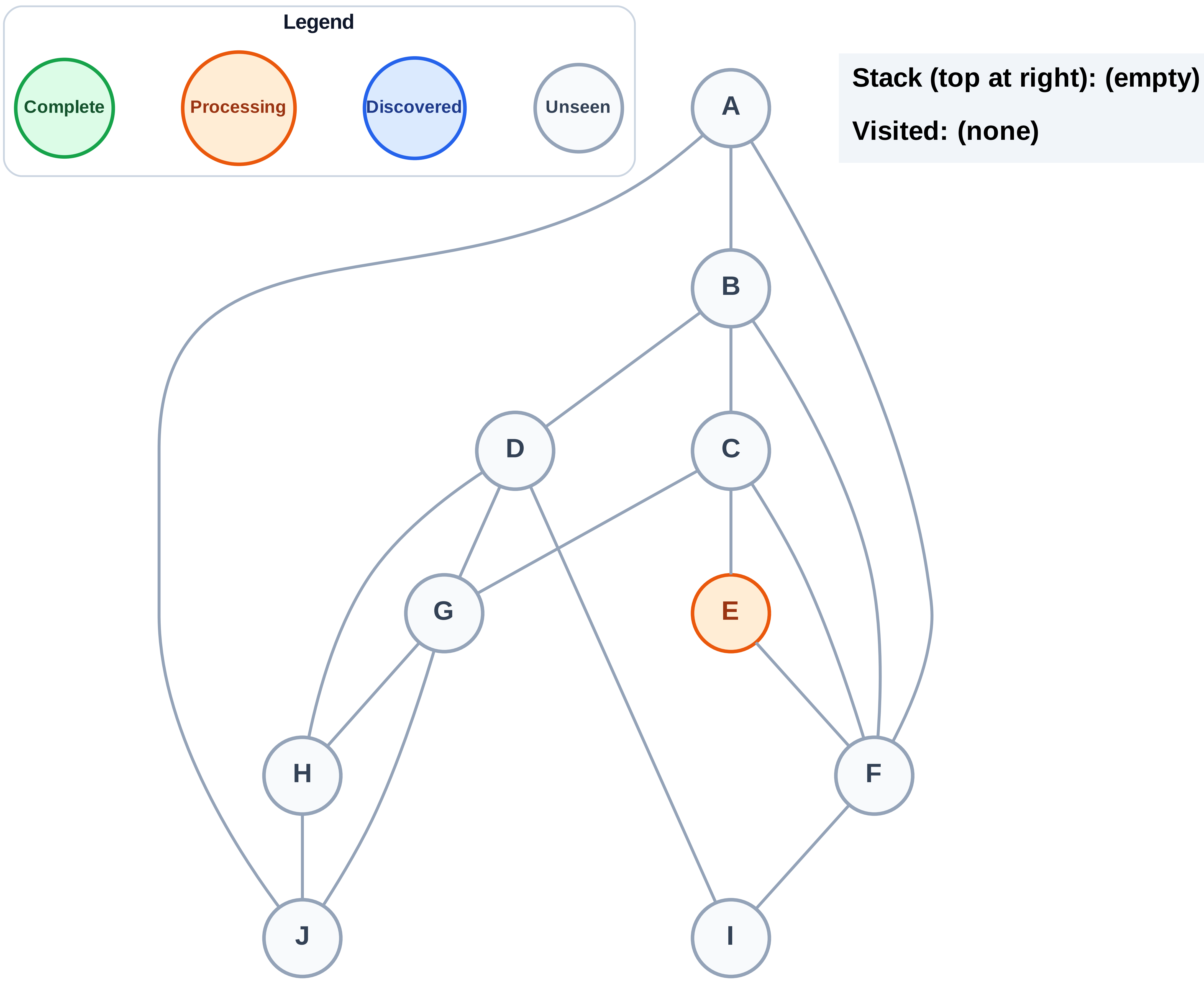
# DFS Traversal

Step 1: Start at E



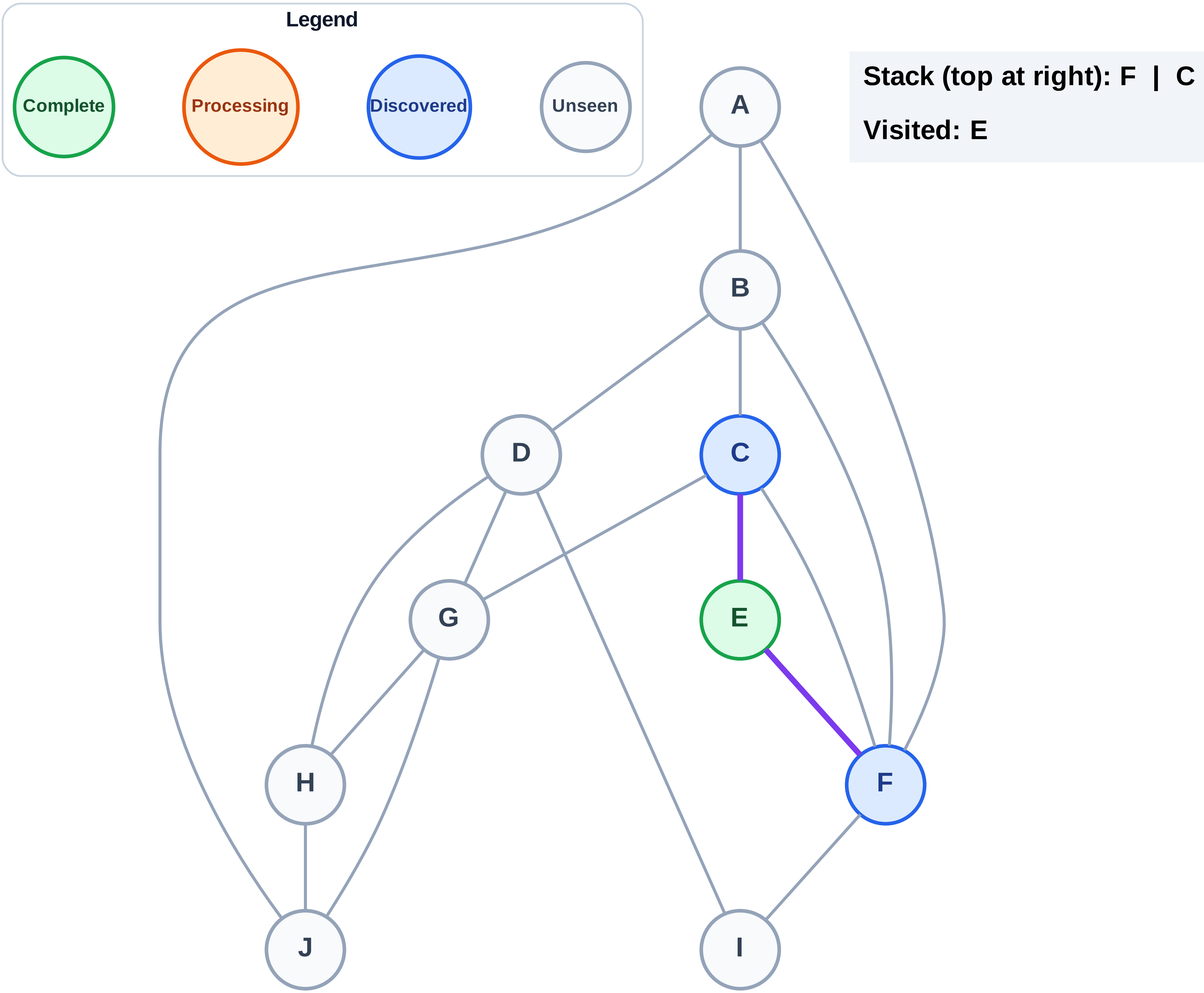
# DFS Traversal

Step 2: Process node E



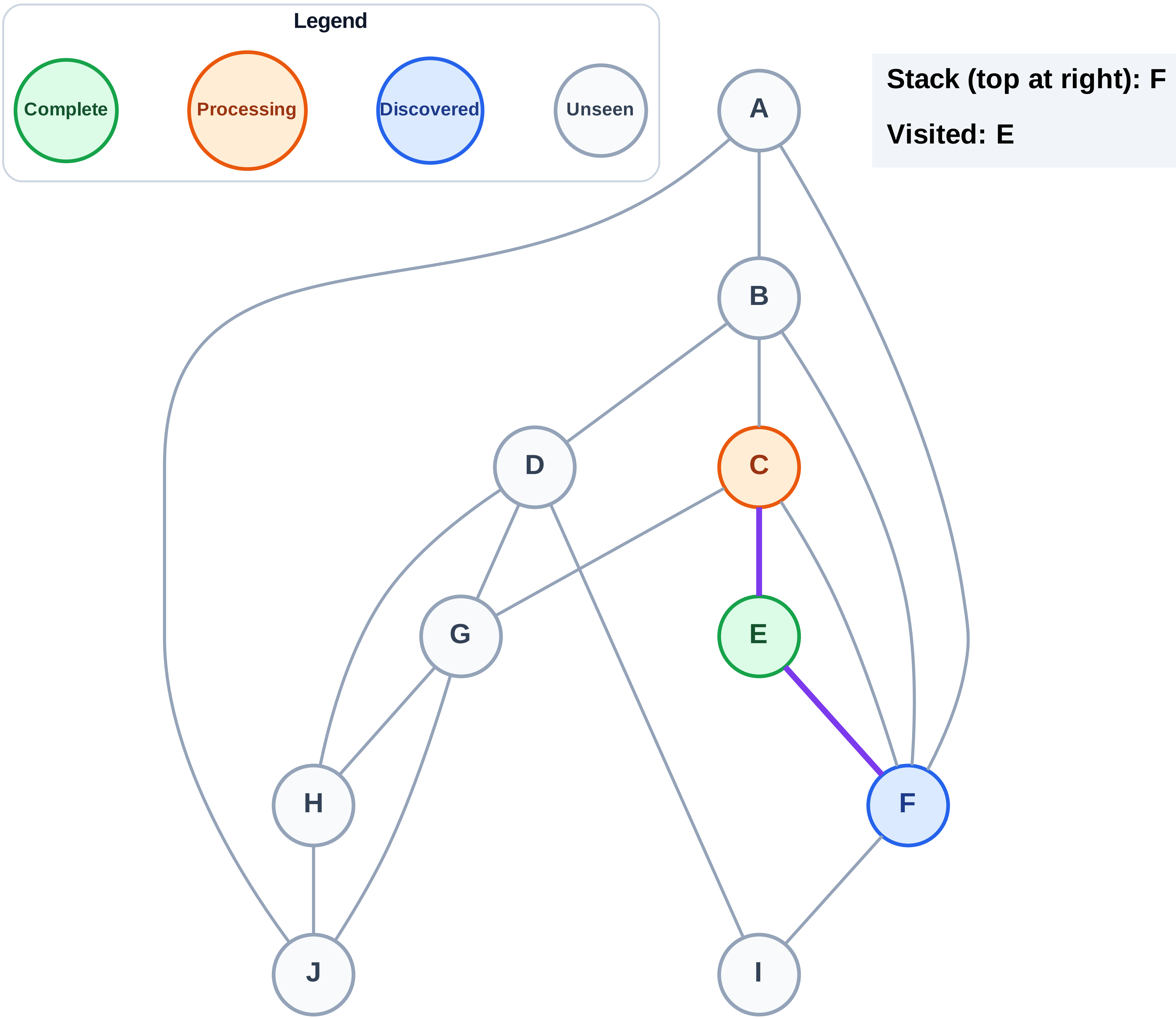
# DFS Traversal

Step 3: Discover F, C from E



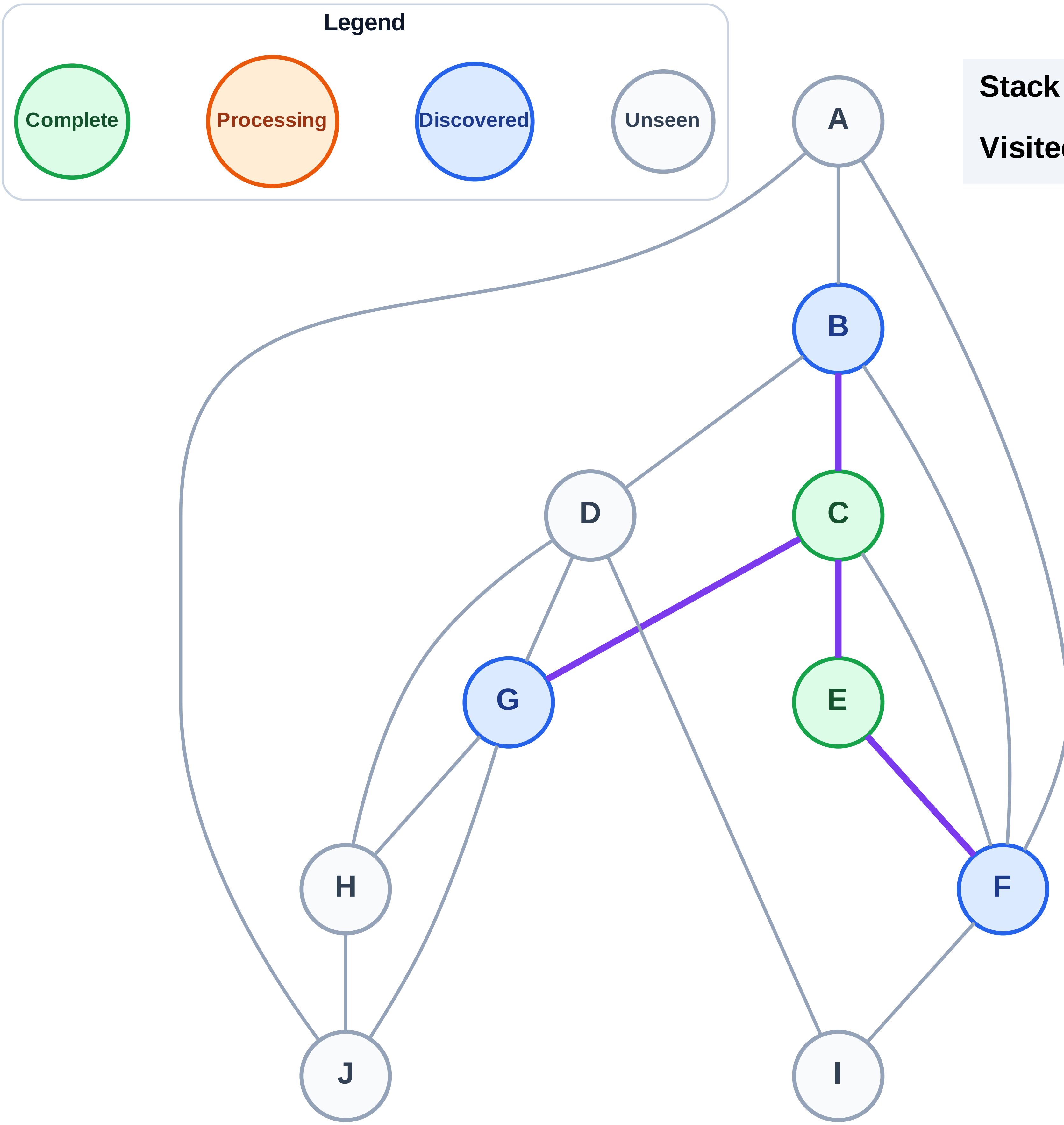
# DFS Traversal

Step 4: Process node C



# DFS Traversal

Step 5: Discover G, B from C

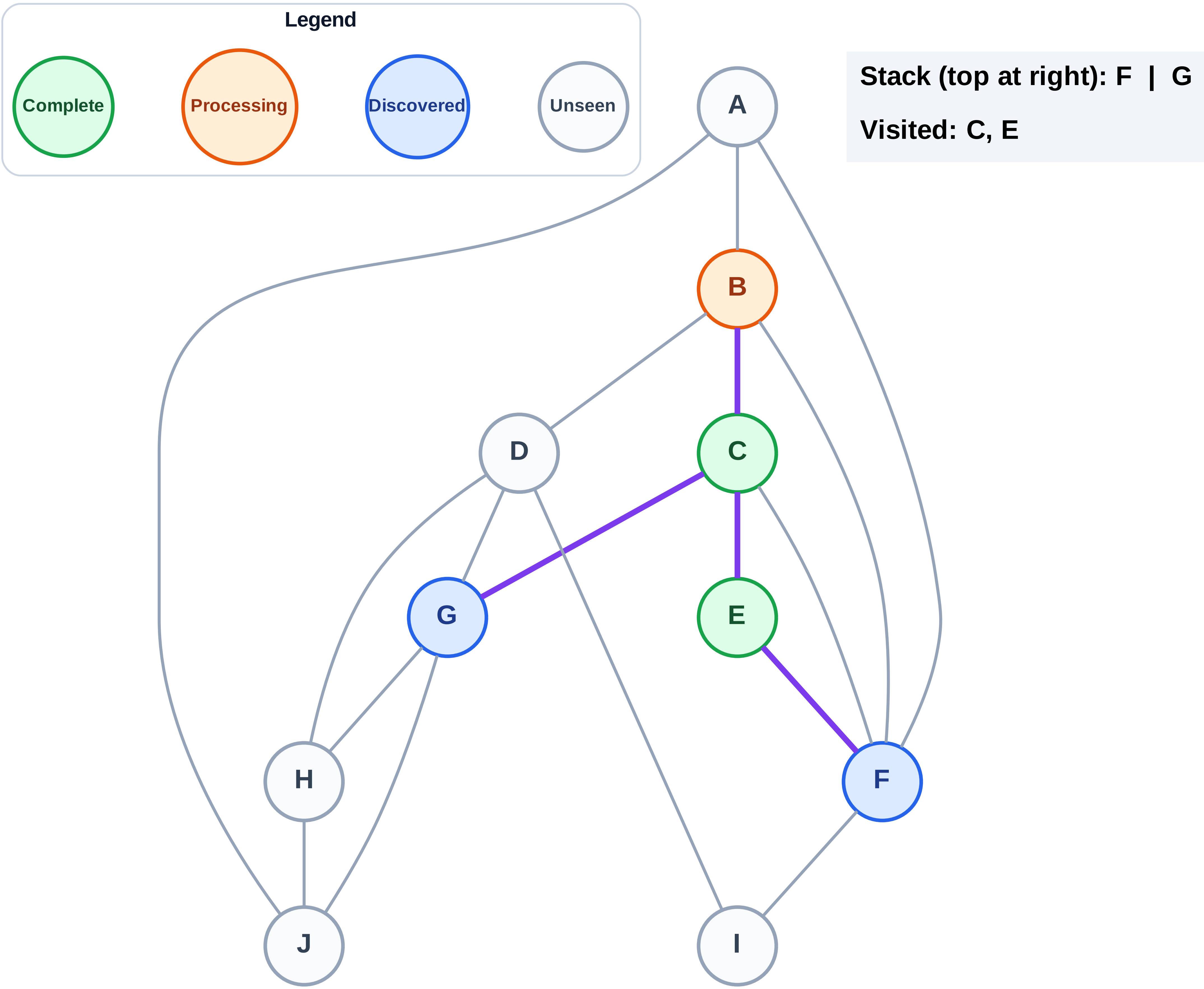


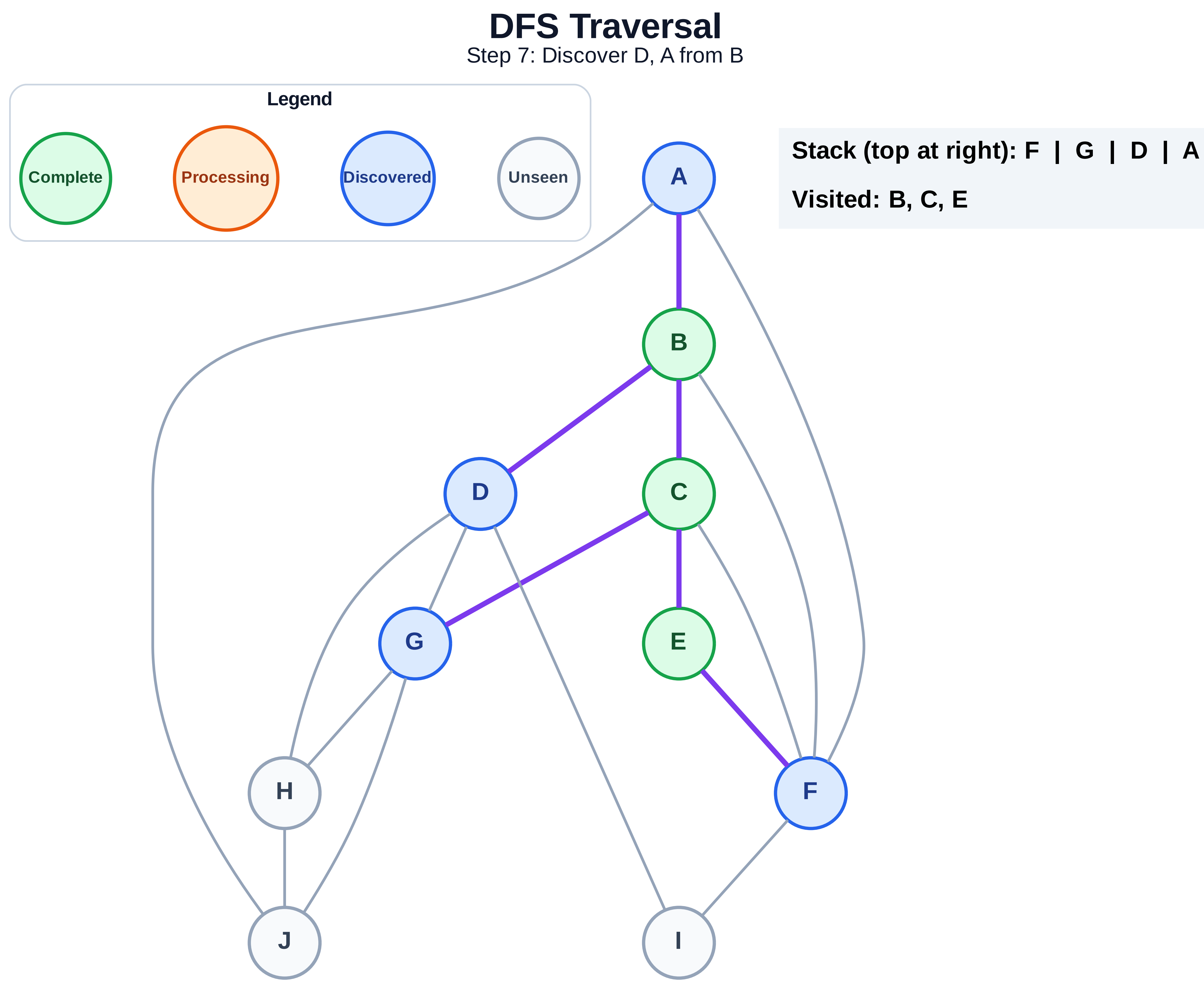

**Stack (top at right): F | G | B**

**Visited: C, E**

# DFS Traversal

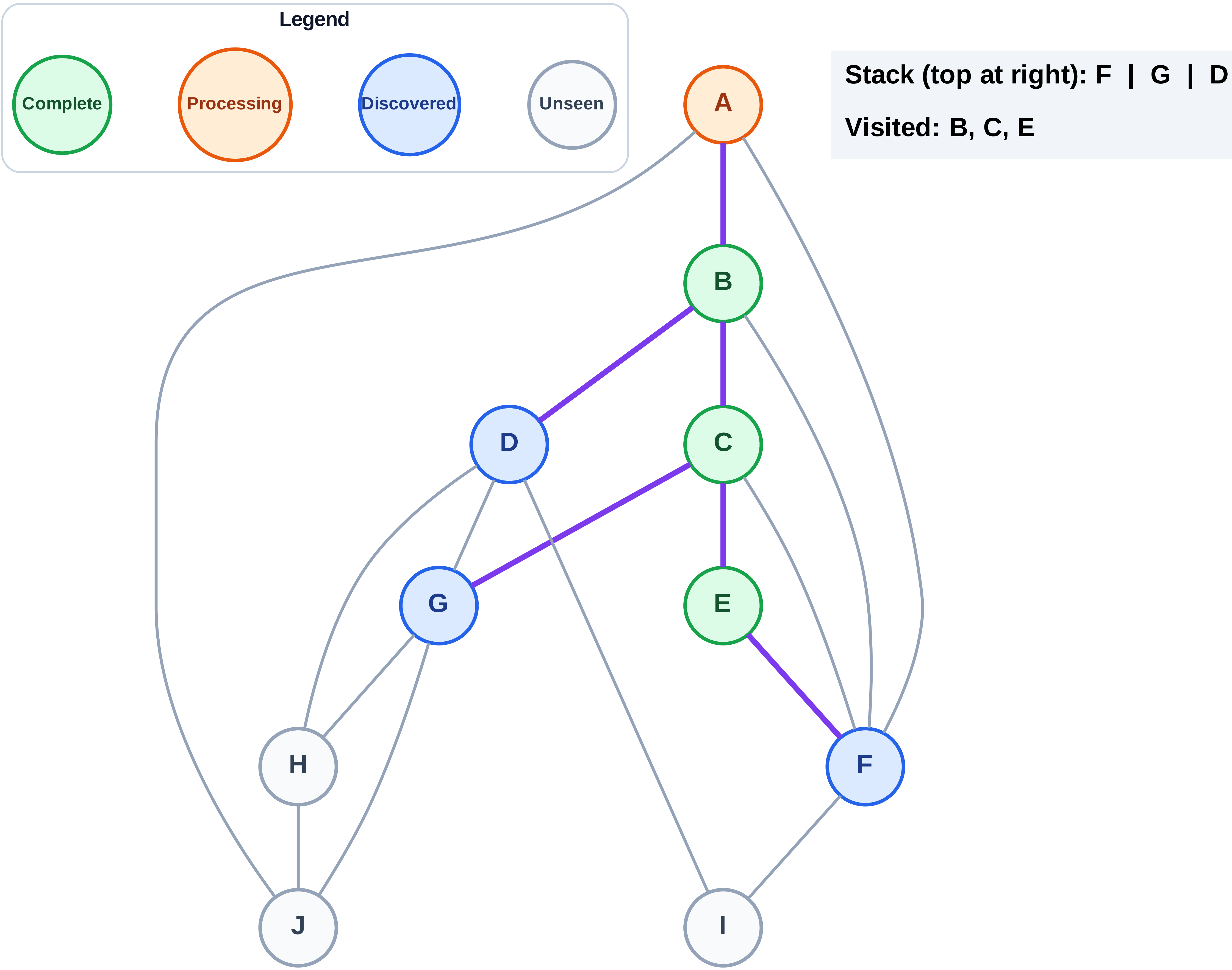
Step 6: Process node B




DFS Traversal
Step 7: Discover D, A from B
Legend
Complete
Processing
Discovered
Unseen
Stack (top at right): F | G | D | A
Visited: B, C, E
A
B
C
D
E
F
G
H
I
J

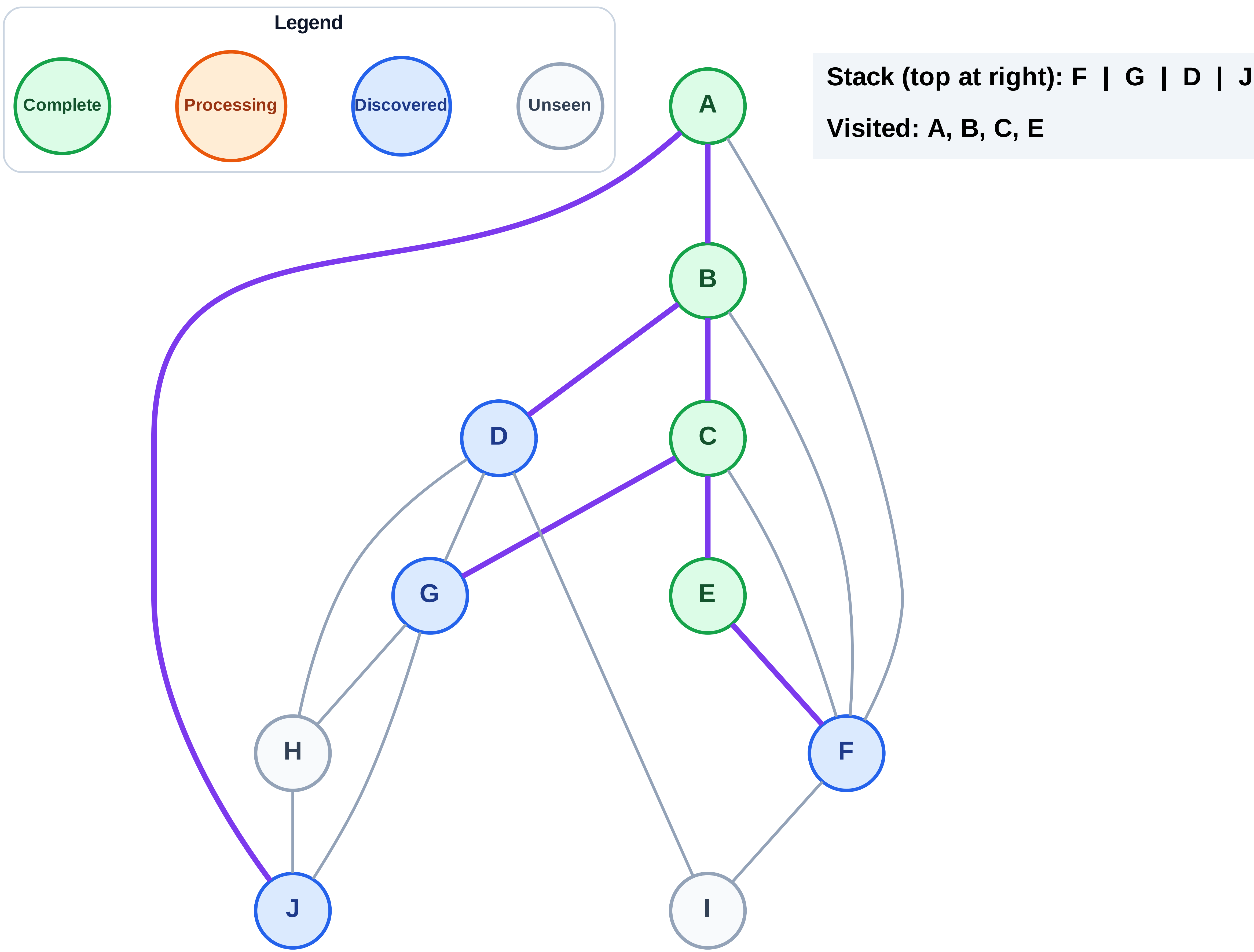
# DFS Traversal

Step 8: Process node A



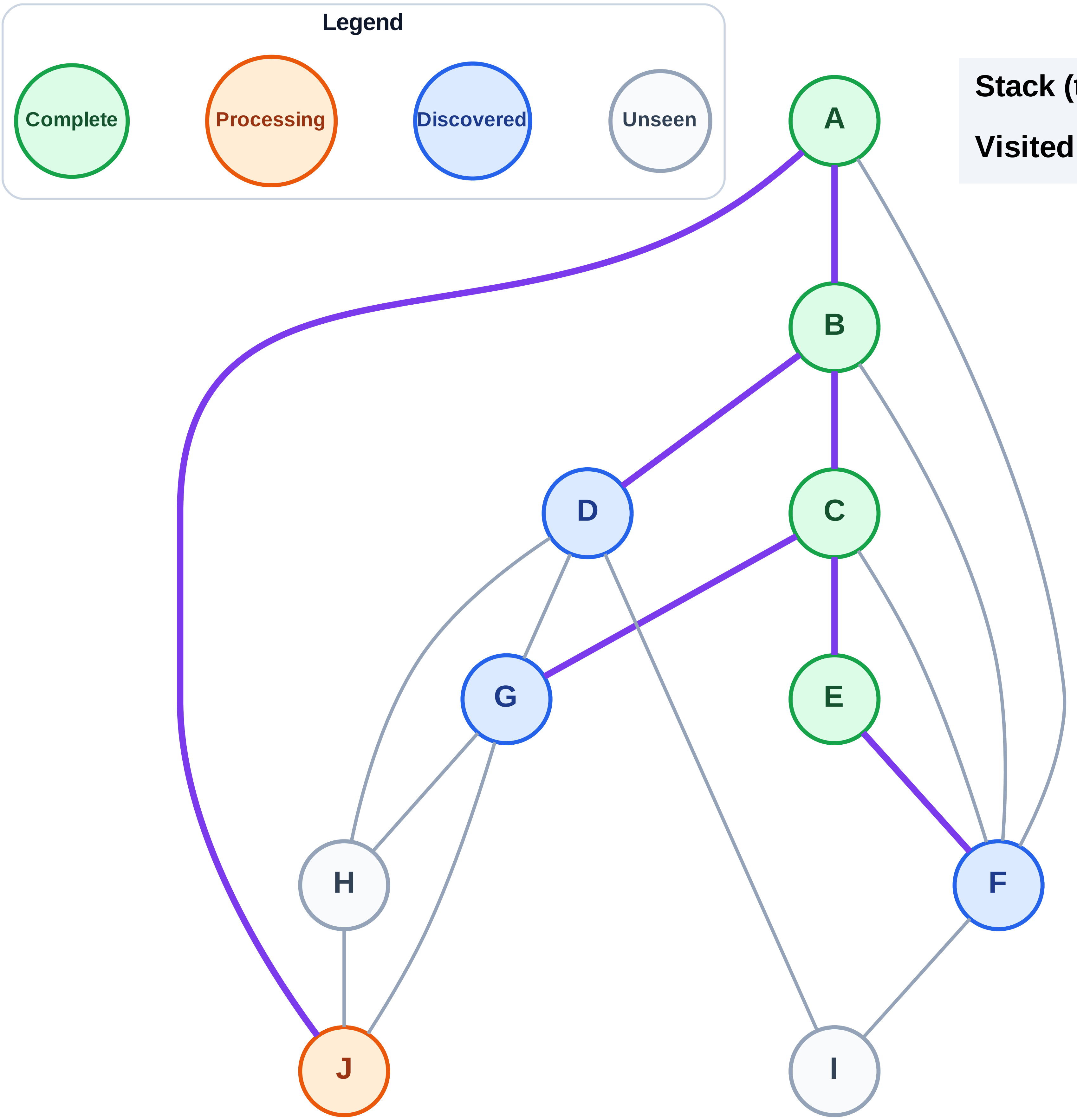
# DFS Traversal

Step 9: Discover J from A



# DFS Traversal

Step 10: Process node J

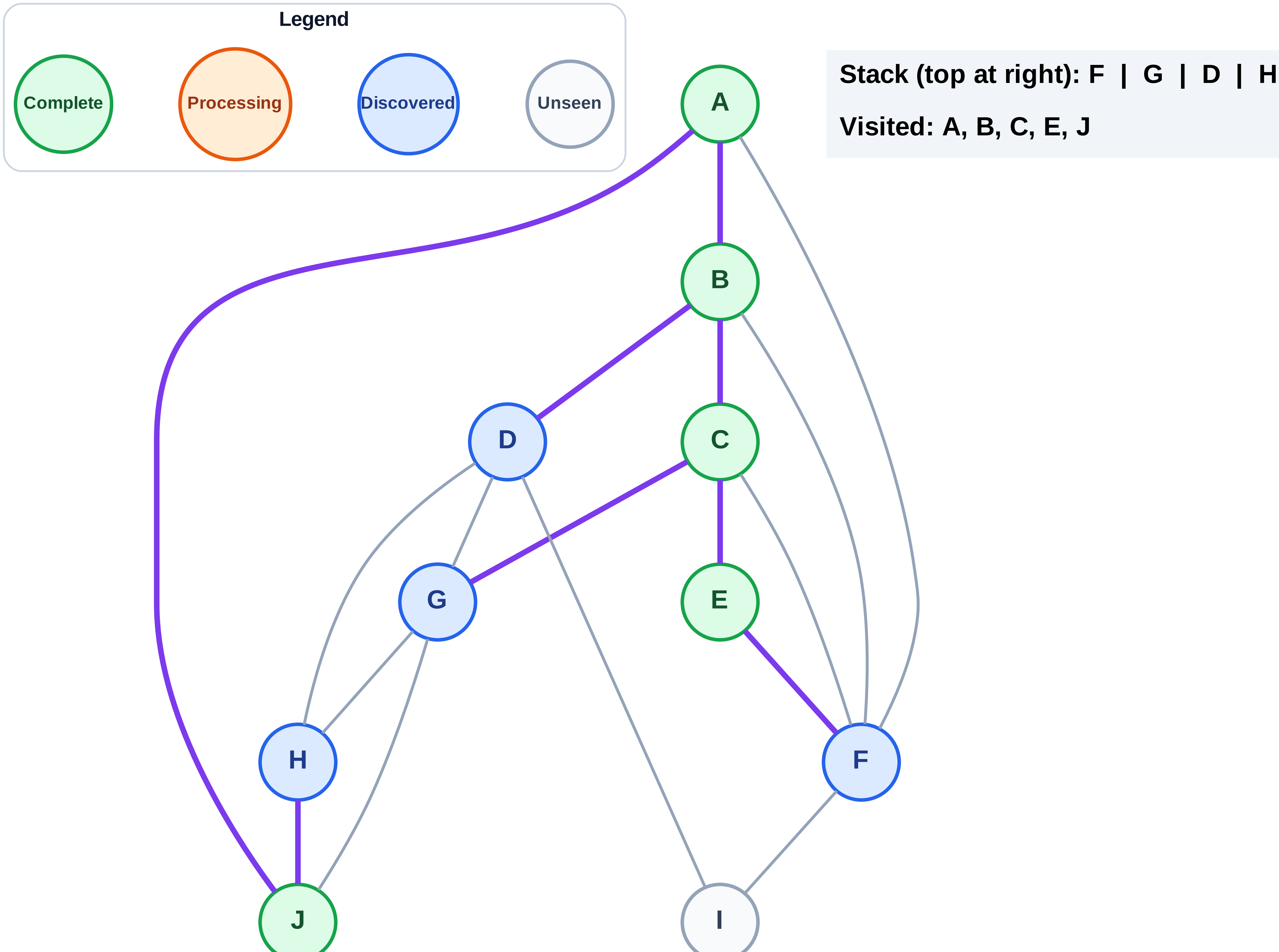



**Stack (top at right): F | G | D**

**Visited: A, B, C, E**

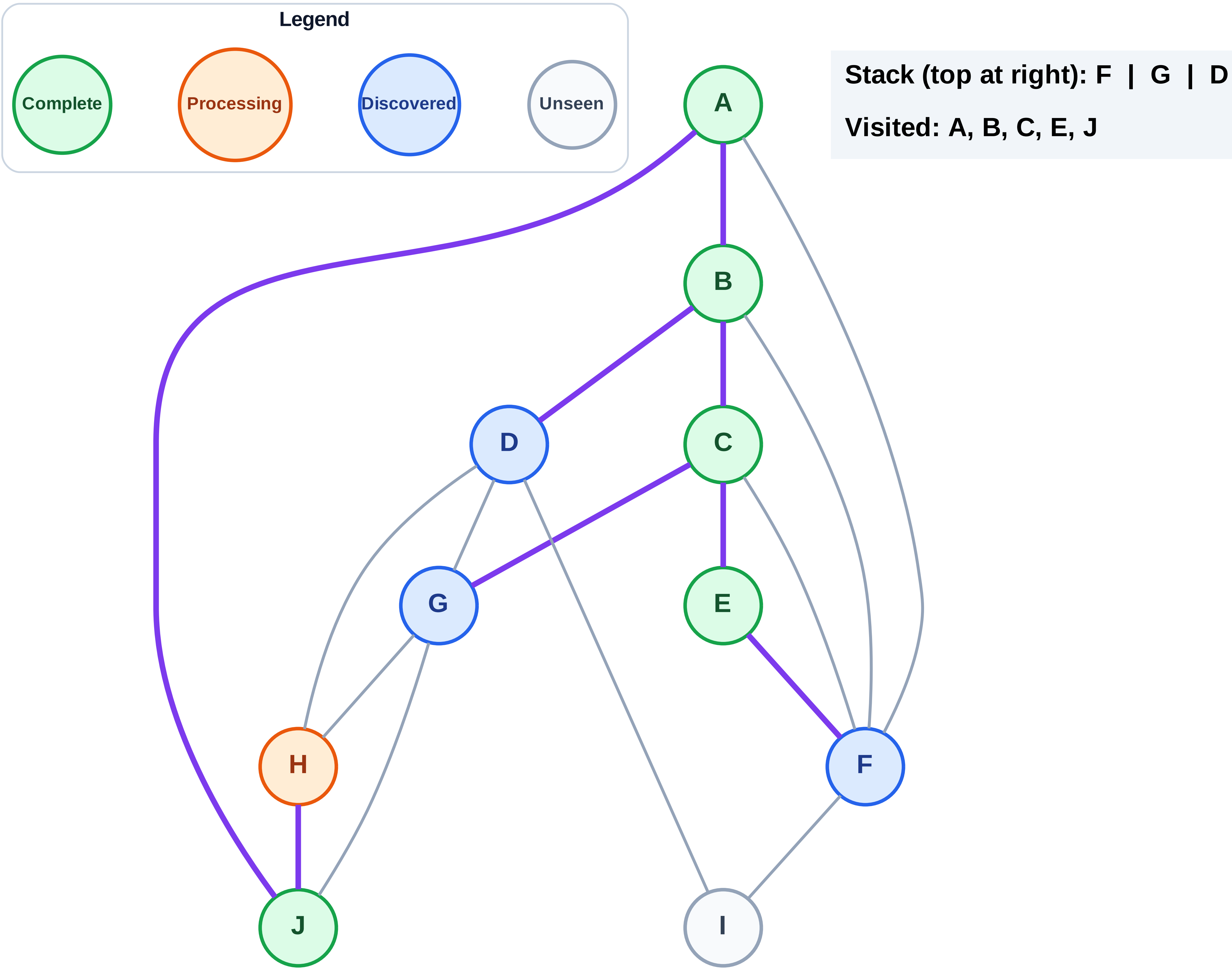
# DFS Traversal

Step 11: Discover H from J



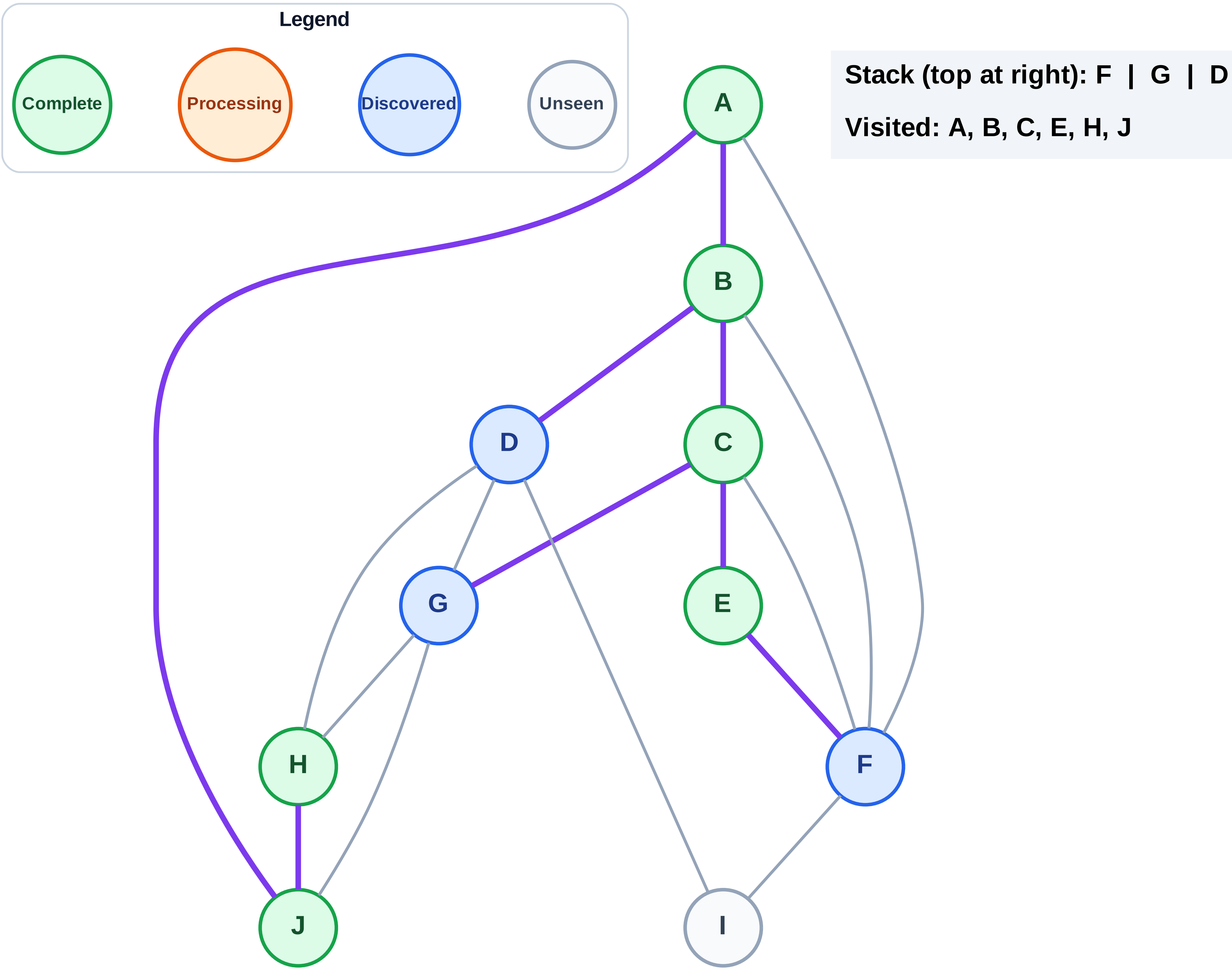
# DFS Traversal

Step 12: Process node H



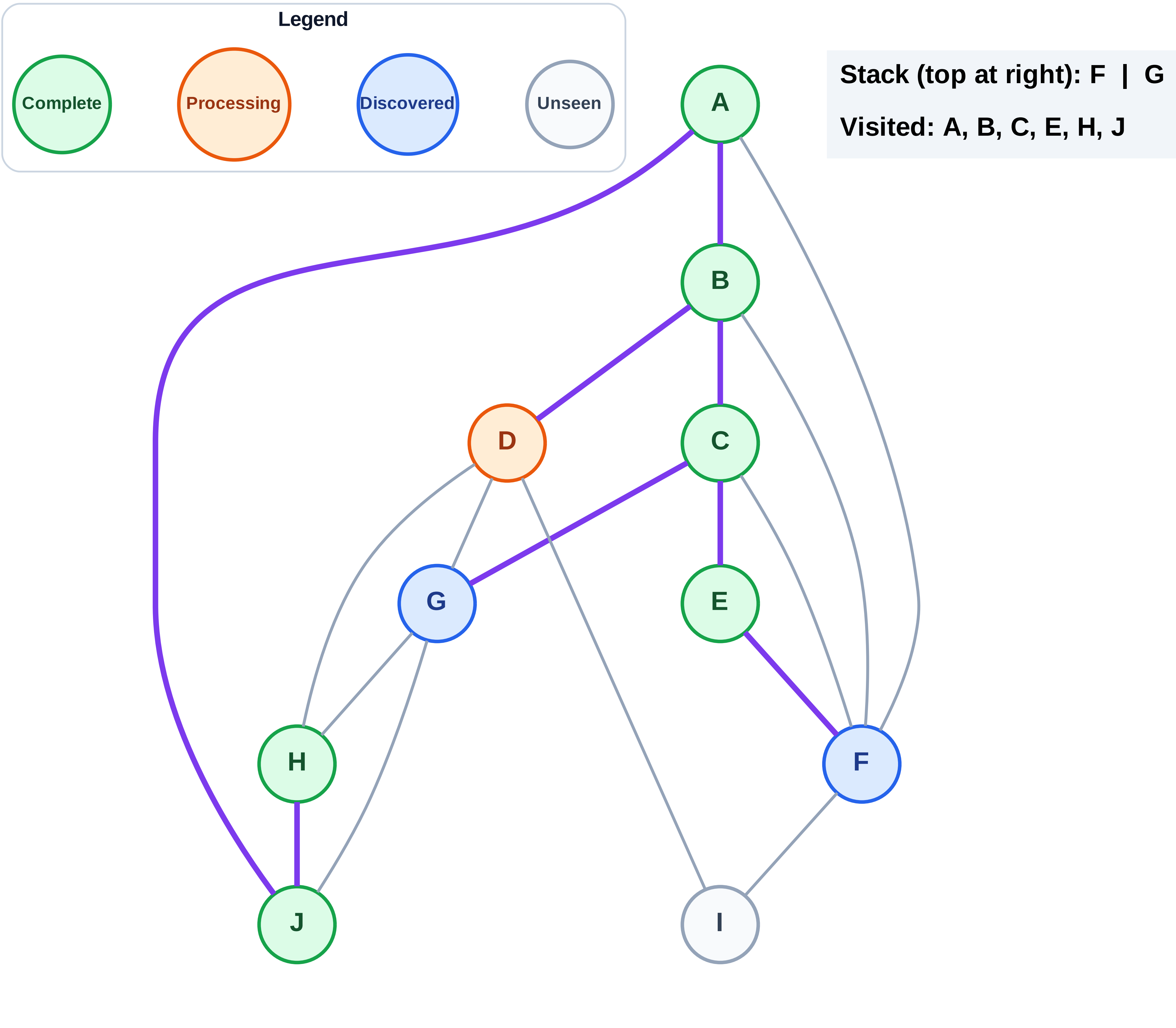
# DFS Traversal

Step 13: No new neighbors from H



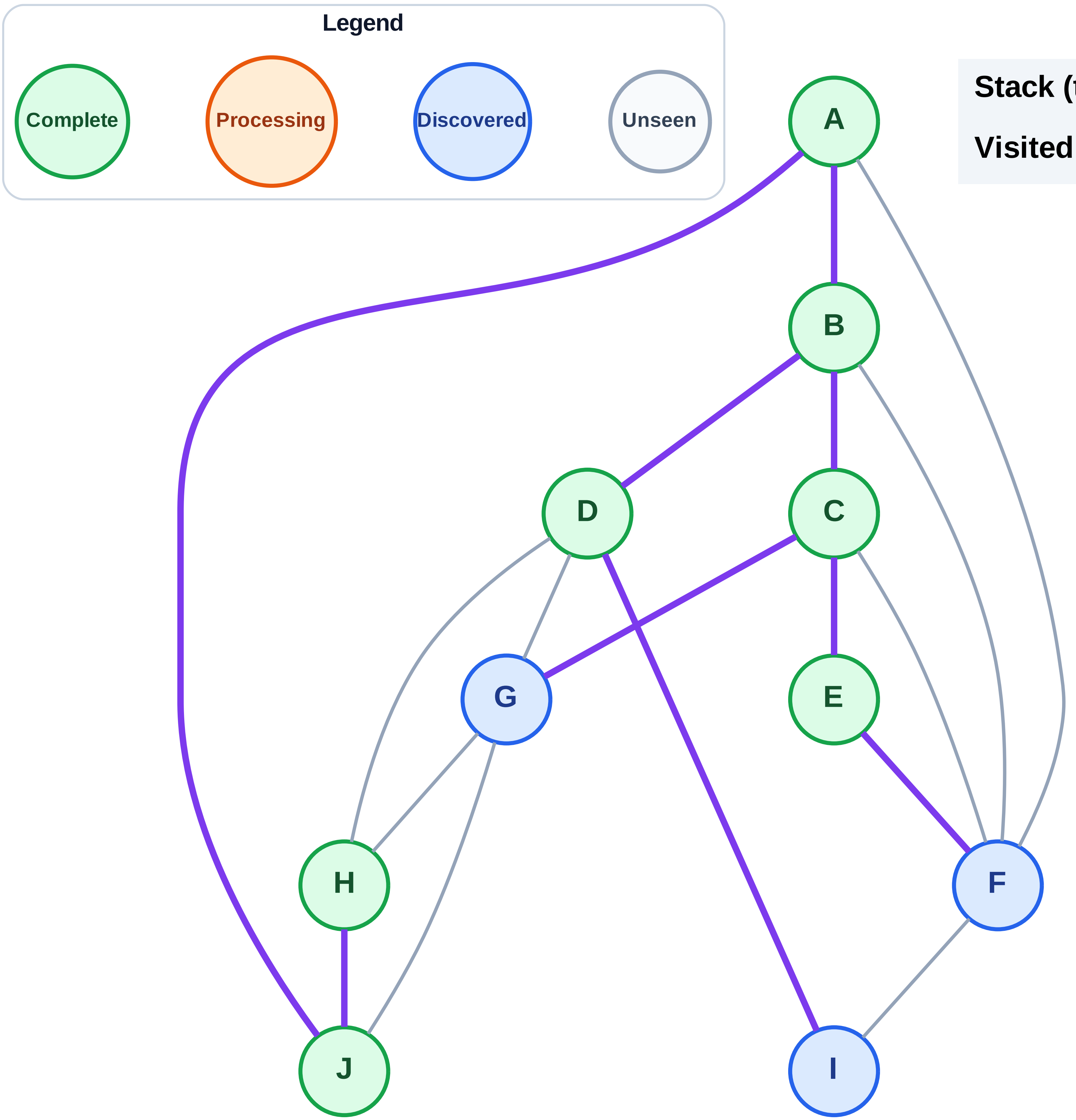
# DFS Traversal

Step 14: Process node D



# DFS Traversal

Step 15: Discover I from D

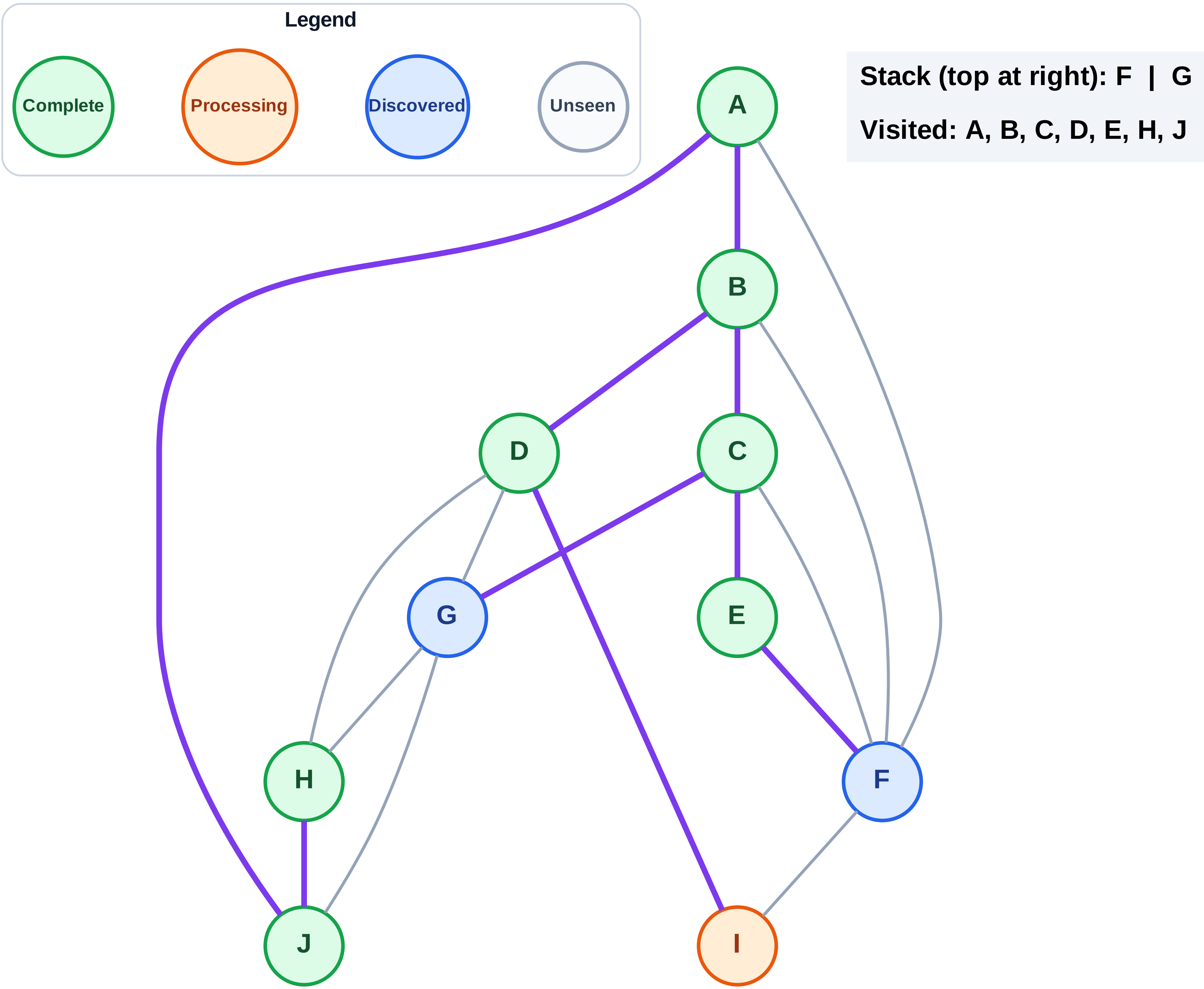



**Stack (top at right): F | G | I**

**Visited: A, B, C, D, E, H, J**

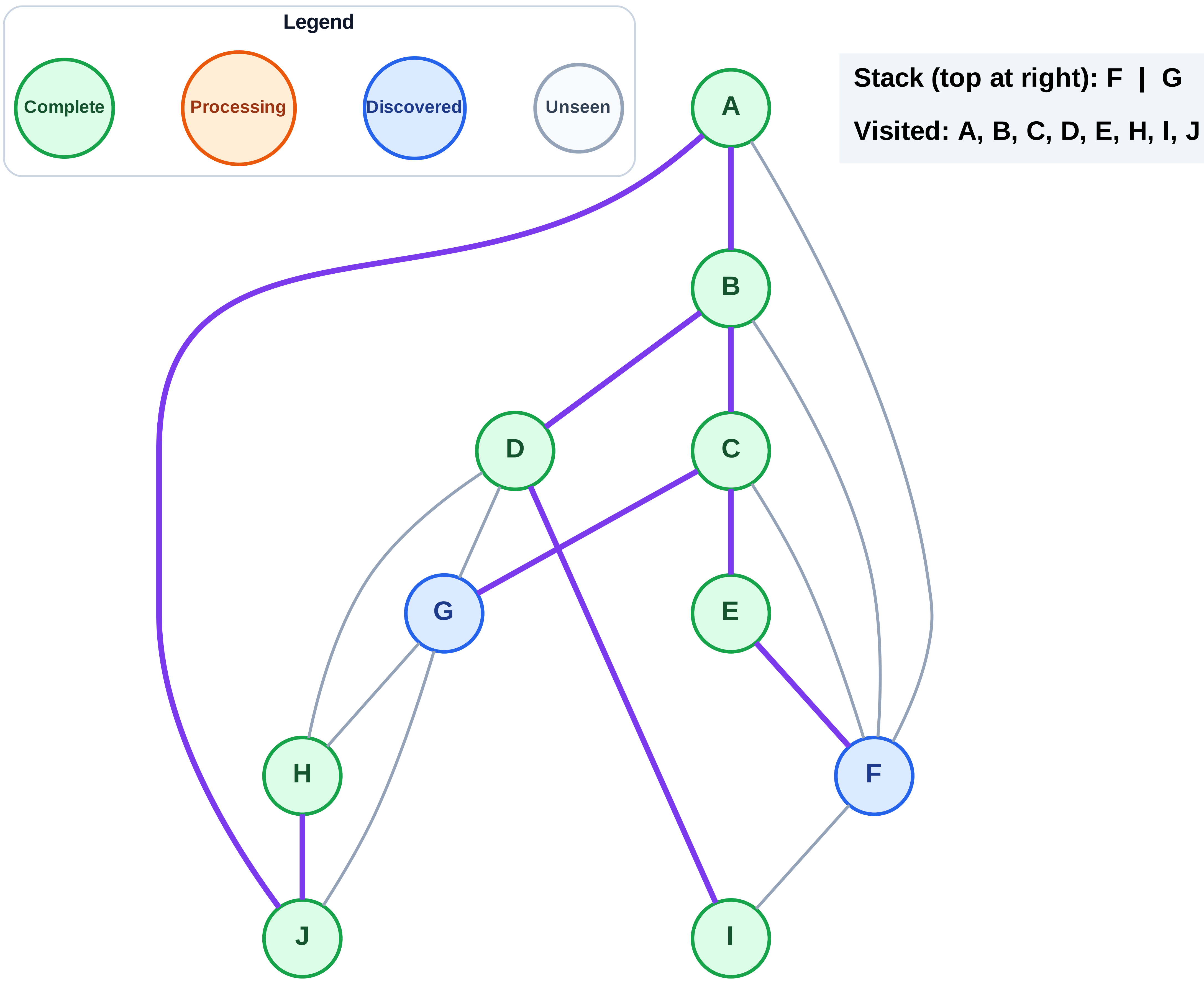
# DFS Traversal

Step 16: Process node I



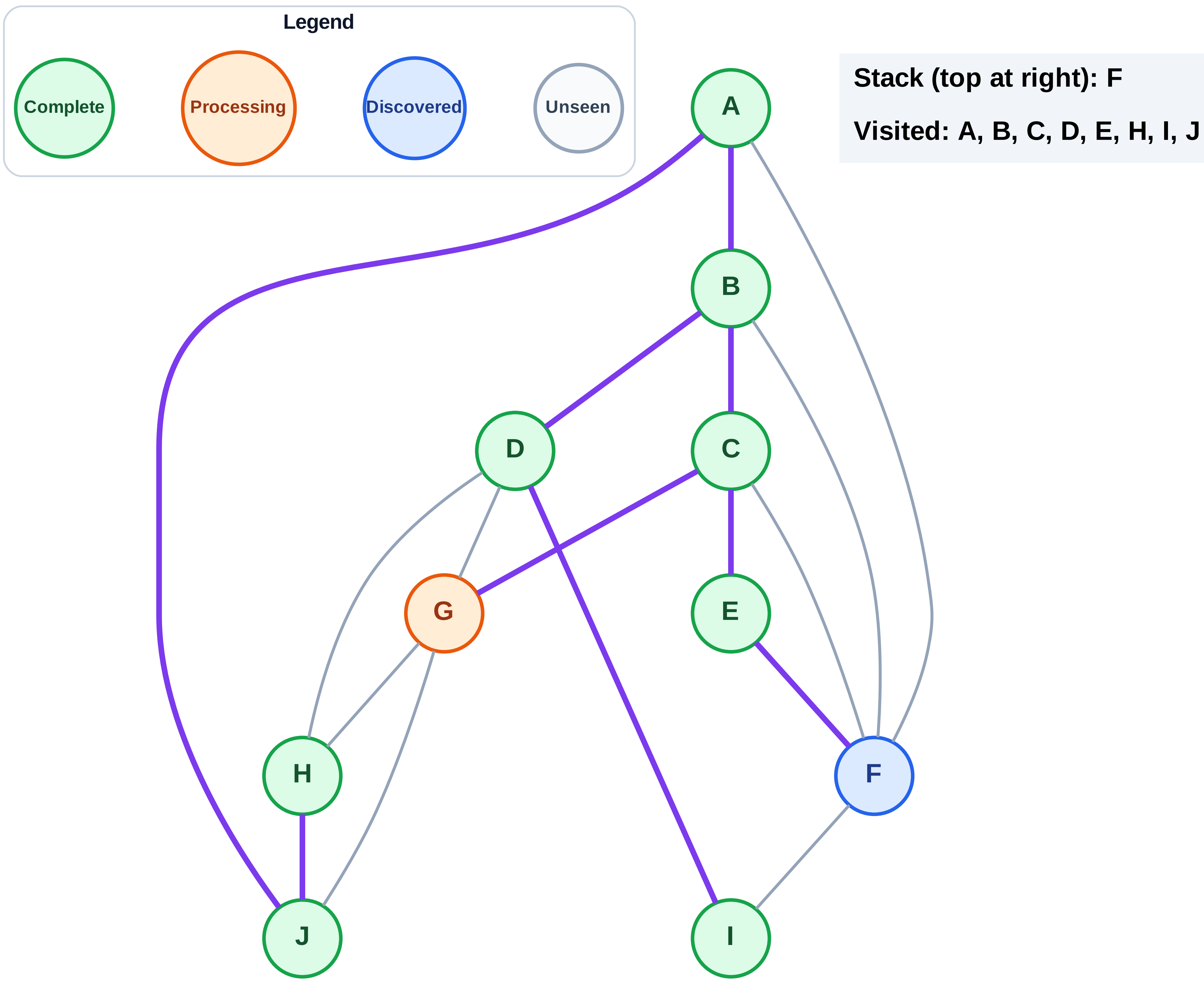
# DFS Traversal

Step 17: No new neighbors from I



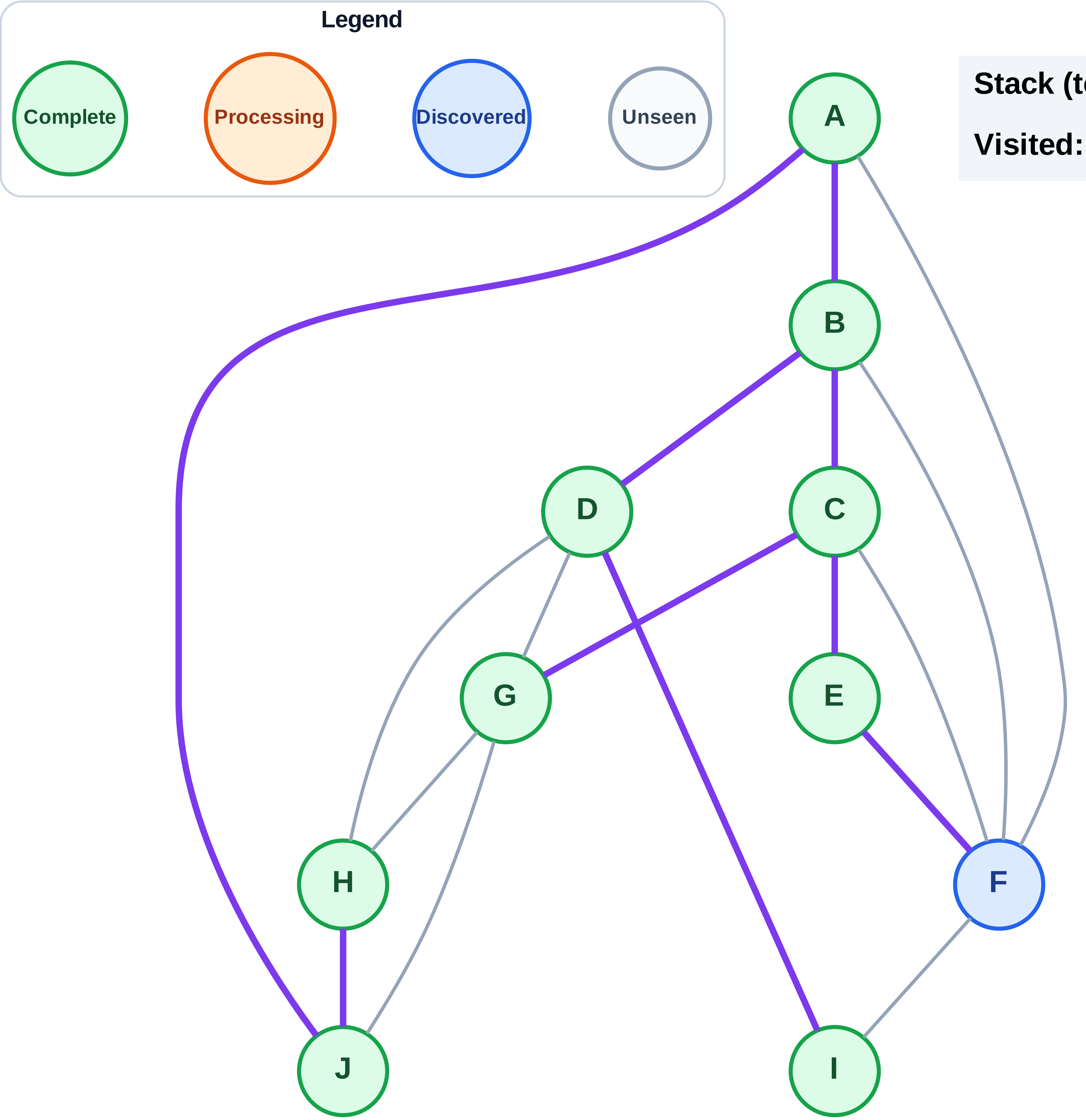
# DFS Traversal

Step 18: Process node G



# DFS Traversal

Step 19: No new neighbors from G

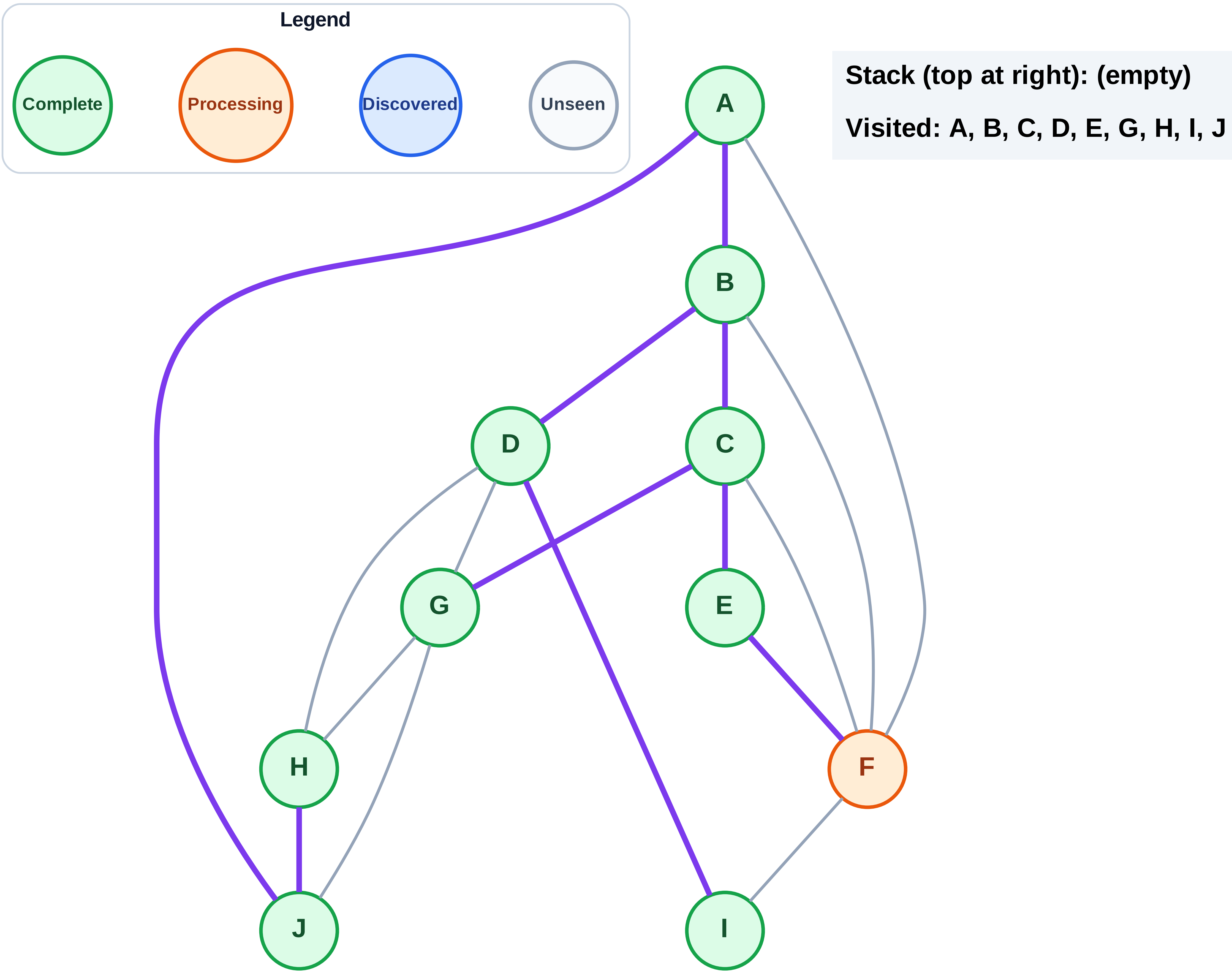



**Stack (top at right): F**

**Visited: A, B, C, D, E, G, H, I, J**

# DFS Traversal

Step 20: Process node F



# DFS Traversal

Step 21: No new neighbors from F